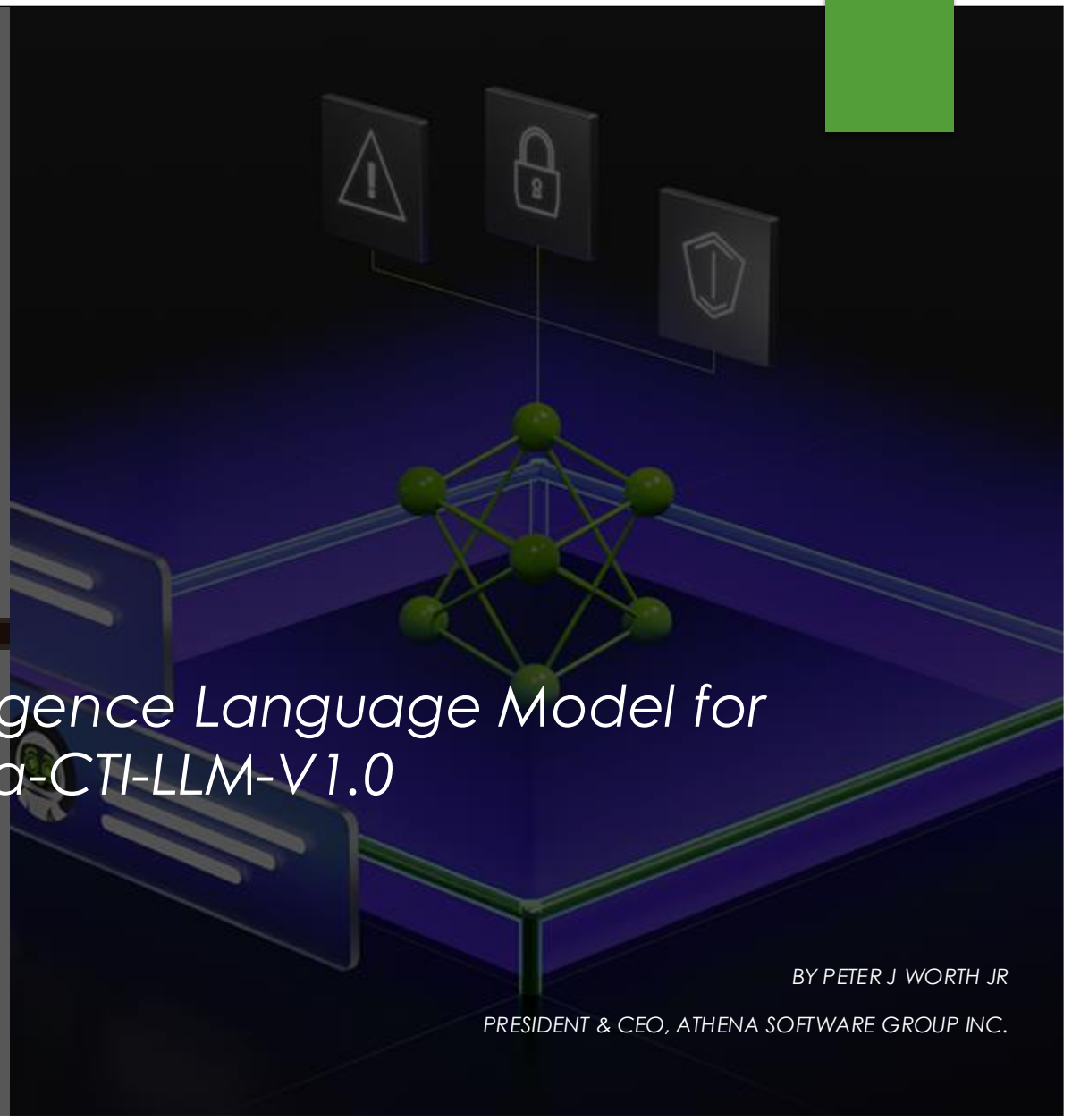
Athena Cyber LLM

Glaukopis

Foundational Cyber Threat Intelligence Language Model for the next generation SOC: Athena-CTI-LLM-V1.0

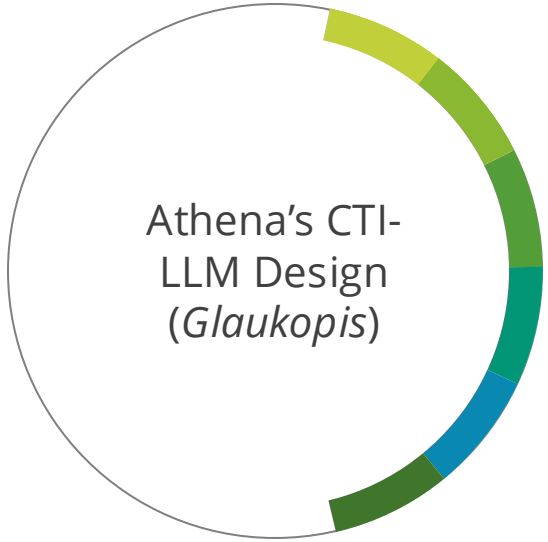


BY PETER J WORTH JR

PRESIDENT & CEO, ATHENA SOFTWARE GROUP INC.

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01 Athena Software Group

02 Cyber LLM Market Overview

03 LLM Finetuning & Optimization

04 Threat Intel Data Integration (Ignite et al)

05 Product Development Lifecycle

06 Summary

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Athena Software Group

WHO WE ARE AND WHY WE ARE DIFFERENT

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Athena Software Group Inc.

A company founded under the principles of integrating Cyber Security best practices, high end custom enterprise software and artificial intelligence (LLMs).

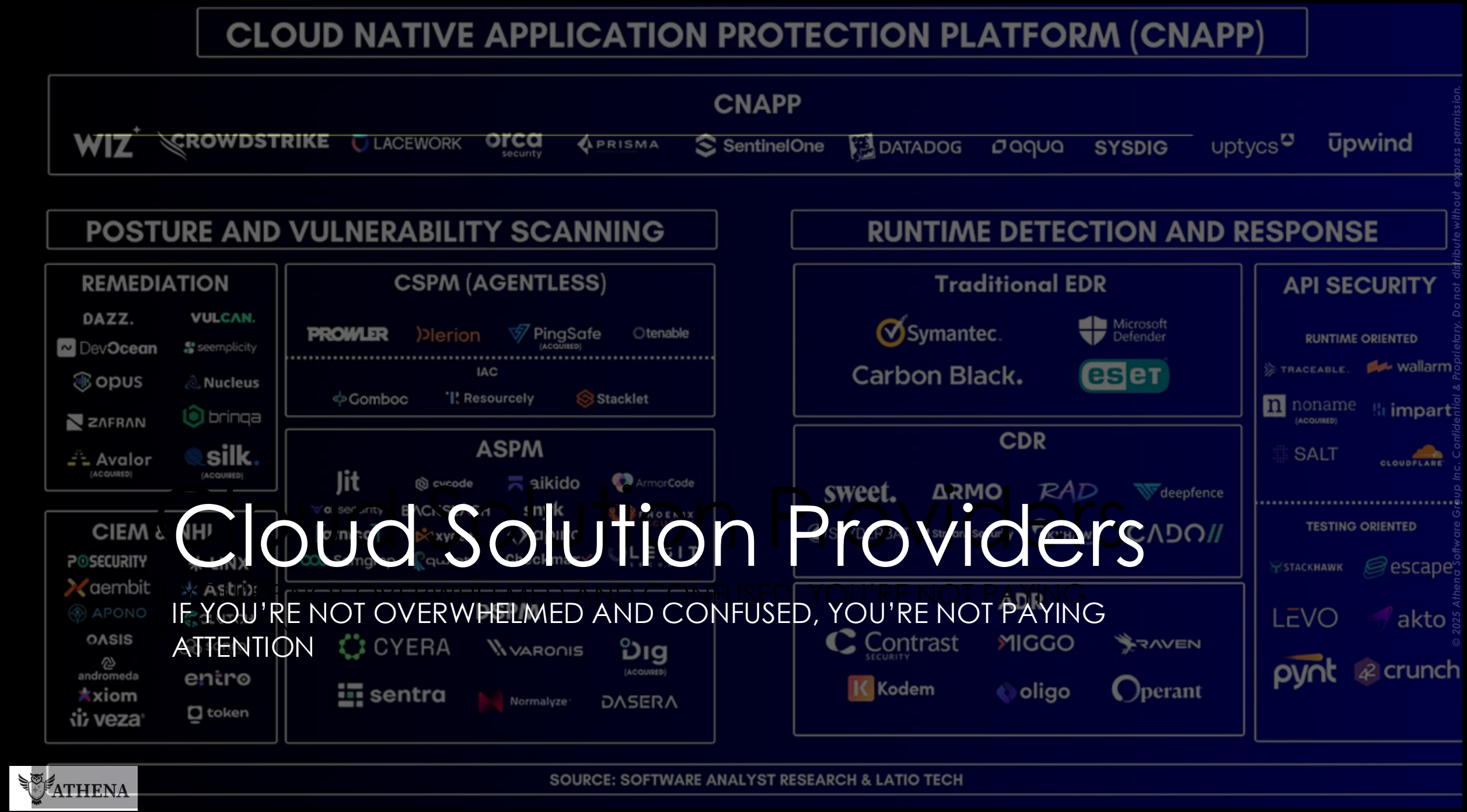
Athena Software Group brings together some of the best minds in computer science, cyber defense and SaaS to design solutions specifically for application service providers.

We take a fully integrated and analytical approach to Cyber Security defense, knowing through experience that the best Cyber Defense strategies arise from providing the best analysts with the best tools from within a fully integrated cyber security infrastructure that is informed by the latest cyber and t

We intend to set a new bar in Cyber defense by providing tools along the entire Cyber defense stack which are fully integrated with Cyber specific trained LLMs that both understand the global threat landscape AND are informed by the technical specifications which underpin the application specific infrastructure environment facilitating higher quality, more rapid and informed outcomes with respect to threat and anomaly detection, response management, vulnerability management and compliance.

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Cloud Solution Providers

IF YOU'RE NOT OVERWHELMED AND CONFUSED, YOU'RE NOT PAYING ATTENTION

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The Cybersecurity Challenge



Proliferation of security tools

Organizations employ a variety of security tools, such as firewalls, web application firewalls (WAFs), intrusion detection systems (IDS), and endpoint detection and response (EDR) solutions, leading to a complex and fragmented security landscape.



Siloed data and lack of integration

Security data is often trapped in individual tools, making it difficult to gain a comprehensive view of the threat landscape and hindering effective threat detection and response.



Overwhelmed security teams

Security teams struggle to correlate and analyze data from multiple disparate tools, leading to increased workloads and delays in identifying and addressing security incidents.

The fragmentation of security tools and siloed data across an organization's network infrastructure hinders the ability to detect, analyze, and respond to threats in a timely and effective manner, presenting a significant challenge in today's evolving cybersecurity landscape.

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The Value of an Integrated Approach

Enhanced Threat Detection

Consolidating security tools and data across the network allows for a more comprehensive view of potential threats, enabling faster and more accurate detection of anomalies and suspicious activities.

Improved Threat Analysis

By integrating data from various security components, security teams can analyze threat patterns and indicators more effectively, leading to better informed decision-making and more efficient incident response.

Streamlined Remediation

An integrated approach facilitates the coordination of remediation efforts, allowing security teams to respond to threats quickly and consistently across the entire network infrastructure.

Reduced Operational Overhead

Consolidating security tools and data eliminates the need for manual correlation and analysis across disparate systems, reducing the time and resources required for security operations.

Enhanced Visibility and Reporting

Integrated security solutions provide a holistic view of the organization's security posture, enabling more comprehensive and meaningful reporting to stakeholders and decision-makers.

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Key Components of Athena’s Fully-Integrated Cyber Defense Solution

Web Application Firewall (WAF)

Protects web applications from common web-based attacks by inspecting and filtering incoming traffic, enforcing security policies, and detecting and mitigating threats such as SQL injection, cross-site scripting, and application-level DDoS attacks.

Firewall

Provides a perimeter defense by controlling and monitoring the incoming and outgoing network traffic based on predetermined security rules, protecting the network from unauthorized access and preventing the spread of malware and other threats.

Network Intrusion Detection System (NIDS)

Monitors network traffic for signs of malicious activity, such as unauthorized access attempts, network-based attacks, and suspicious data patterns, and alerts security teams to potential threats in real-time.

Endpoint Detection and Response (EDR)

Continuously monitors and analyzes activity on individual endpoints (e.g., laptops, desktops, servers) to detect, investigate, and respond to advanced threats, including malware, zero-day attacks, and suspicious user behavior.

Security Information and Event Management (SIEM)

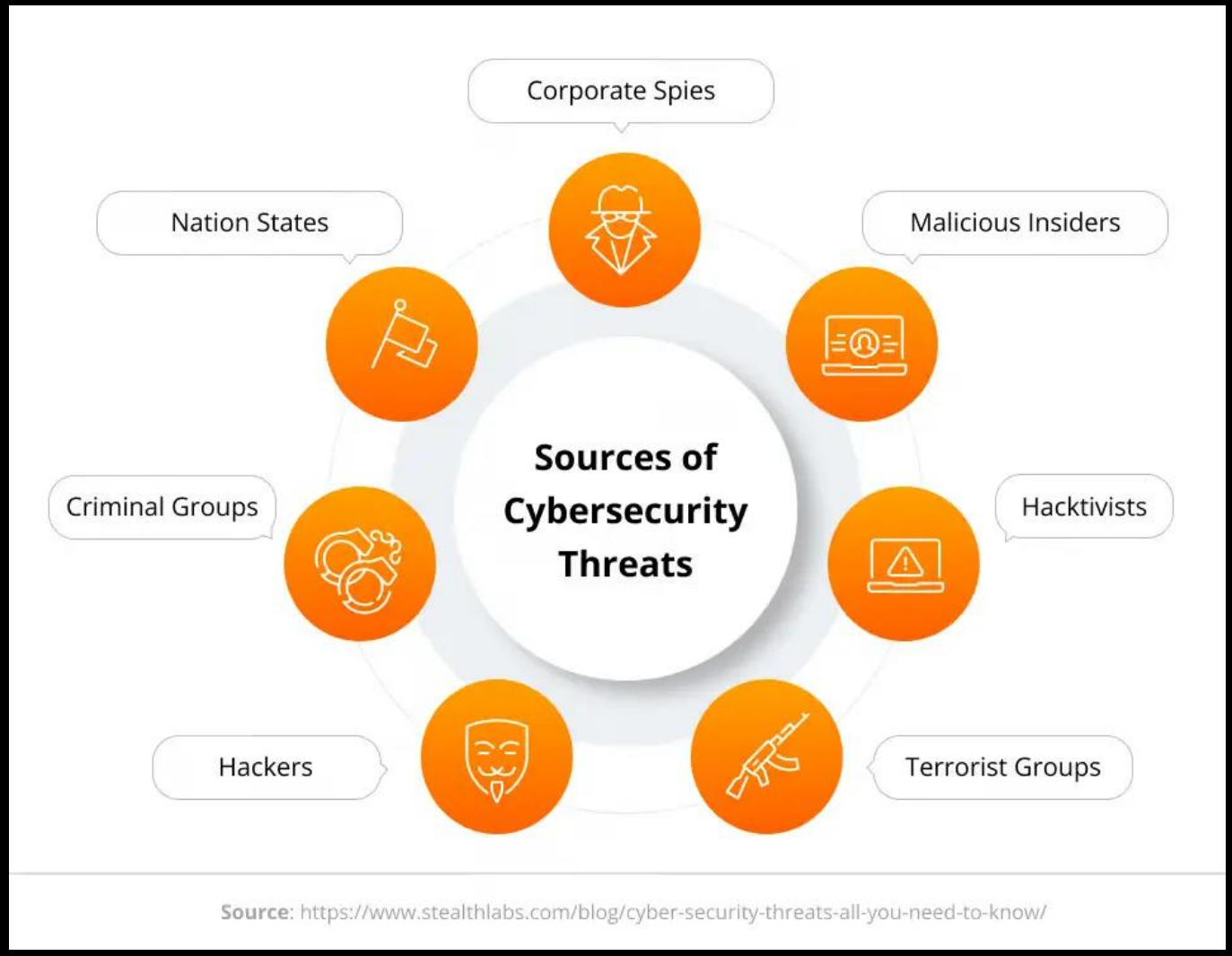
Collects, analyzes, and correlates security-related data from various sources across the network, providing a centralized view of the organization's security posture and enabling faster detection, investigation, and response to security incidents.

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Maximizing Cybersecurity with an Integrated Defense Strategy

By adopting an integrated cyber defense strategy that consolidates security tools and data across the network, organizations can significantly improve their ability to detect, analyze, and respond to threats in a timely and effective manner. This approach not only enhances overall cybersecurity posture but also empowers security teams to make more informed decisions and optimize their resources.



Source: https://www.stealthlabs.com/blog/cyber-security-threats-all-you-need-to-know/

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Athena
the Product Suite

Athena Software Group was founded with the vision that it should not be complicated to setup and maintain a best in class Cyber security posture for an enterprise application. The idea originates from the experience of one of the founders in building and supporting large scale, secure SaaS application infrastructures that house protected data for over two decades.

Most security software solutions are targeted at the entire corporate network infrastructure and as such CISO's and Security Operations professionals who are responsible for maintaining the security posture of ASPs are left with the task of cobbling together a variety of tools across the observability, telemetry, IDS, SIEM, WAF and performance monitoring landscape to gain full visibility into their environment and detect and prevent threats.

This problem is further exacerbated with the deluge of information in the form of reports, email alerts and SIEM data that comes from these disparate providers, creating a complex and noisy monitoring and alerting environment where the discernment of critical information itself becomes a challenge, affecting incident response times which is the most critical metric in determining the extent and scope, and ultimately cost, of a data breach.



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Glaukopis (γλαυκῶπις)

the "grey eyed one"





Here's a more detailed breakdown:

"Glaukós" (γλαυκός):

This word, meaning "gleaming, silvery," is the root of the name. It later came to be associated with the color bluish-green or grey.

"Ōps" (ὤψ):

This word means "eye".

"Glaux" (γλαῦς):

This is the Greek word for "owl," specifically the small owl, Athene noctua (Latin).

"Glaukopis" (γλαυκῶπις):

The name, combining "glaukós" and "ṓps," is often translated as "owl-eyed," "bright-eyed," or "grey-eyed," depending on the interpretation.

The name's connection to owls likely stemmed from Athena's association with wisdom and owls, who are often seen as symbols of intelligence and knowledge. The owl's eyes, particularly their piercing gaze, may have inspired the description of Athena as "owl-eyed" or "gleaming-eyed".

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Our Services



Core Cyber Security Tools (Promachos)

Subscription fees, sized according to application complexity & scale



Threat Intelligence Model (Glaukopis)

Custom Threat Intelligence LLM, license model channel dependent



Cyber Product Development & Operations Support

Outsourced cyber security related product and/or operational support. Priced per project.

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Product Roadmap & Go to Market **Timeline**

2025

ALPHA

- Entity & Team Build.
- Site setup and Branding
- Proof of Concept
- Go to Market Strategy
- Funding

2026

BETA

- Beta Release
- Production Operations
- Design Partners & Solution Delivery
- Product Refinement & optimization

2027

GAMMA

- GA
- SOC Operations & outsourcing
- Partner Channels
- Sales & Marketing Scale Additional Funding, Valuation

2028

Delta

- Market Disruption
- Generation II LLM
- Larger Customer Deployments
- Custom Engagements

- ✓ *Release Early & Often*
- ✓ *Engage Design Partners Early for market fit and revenue alignment*
- ✓ *Raise funds only as necessary*

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Cyber Threat Intel Trained LLMs (CTI-LLM)

USE CASE MODEL, DOMAIN OVERVIEW

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Foundational Use Cases for Cyber Security Language Models

1. Threat Detection and Anomaly Identification

LLMs can analyze extensive logs, network events, and telemetry data, recognizing subtle patterns indicative of cyber threats or anomalies. They provide contextual understanding of suspicious activity, correlating seemingly unrelated events to identify potential breaches or attacks.

2. Security Incident Response Automation

Automating responses to standard security incidents by interpreting alerts, analyzing context, and suggesting or executing initial remediation steps. Streamlining the triage process to quickly prioritize critical threats based on context derived from past incidents.

3. Vulnerability Assessment and Management

Parsing security bulletins, advisories, and CVE databases to summarize and highlight critical vulnerabilities relevant to an organization's infrastructure. Generating actionable mitigation strategies and prioritization recommendations tailored to the organization's specific technologies and risk profile.

4. Phishing and Social Engineering Detection

Identifying suspicious patterns in emails, text messages, and other communication platforms by leveraging natural language understanding to detect phishing, impersonation, or manipulative techniques. Educating users by generating realistic phishing simulations or context-aware cybersecurity training content.

5. Cybersecurity Documentation and Compliance Automation

Generating compliance reports, incident documentation, and audit logs automatically, reducing the manual burden associated with regulatory requirements (e.g., GDPR, NYSDFS 23 NYCRR 500).Assisting in the interpretation of regulatory texts, summarizing complex cybersecurity requirements, and suggesting compliance actions.

6. Intelligent Threat Intelligence

Parsing vast amounts of threat intelligence data to provide concise, actionable insights tailored to an organization's unique threat landscape. Automatically generating intelligence reports or briefings about emerging threats and their implications for the organization.

7. Cybersecurity Assistant (Chatbot/Virtual Analyst)

Serving as a first-line assistant that provides cybersecurity support, answering questions about policies, procedures, recent threats, or incident response guidelines. Assisting cybersecurity analysts by summarizing logs, incidents, or threat intelligence data to expedite informed decision-making.

8. Code and Software Security Analysis

Reviewing code repositories, documentation, and logs to identify potential security weaknesses, such as insecure coding patterns, hardcoded credentials, or risky configuration settings. Suggesting secure coding practices and generating contextual remediation steps.

9. Enhanced SIEM and Log Analysis

Integrating LLMs with Security Information and Event Management (SIEM) platforms to provide enhanced semantic understanding of security logs, correlate events, and reduce false positives through natural language analysis. Improving root-cause analysis by interpreting event logs and contextualizing security events.

10. Retrieval-Augmented Generation (RAG) for Security

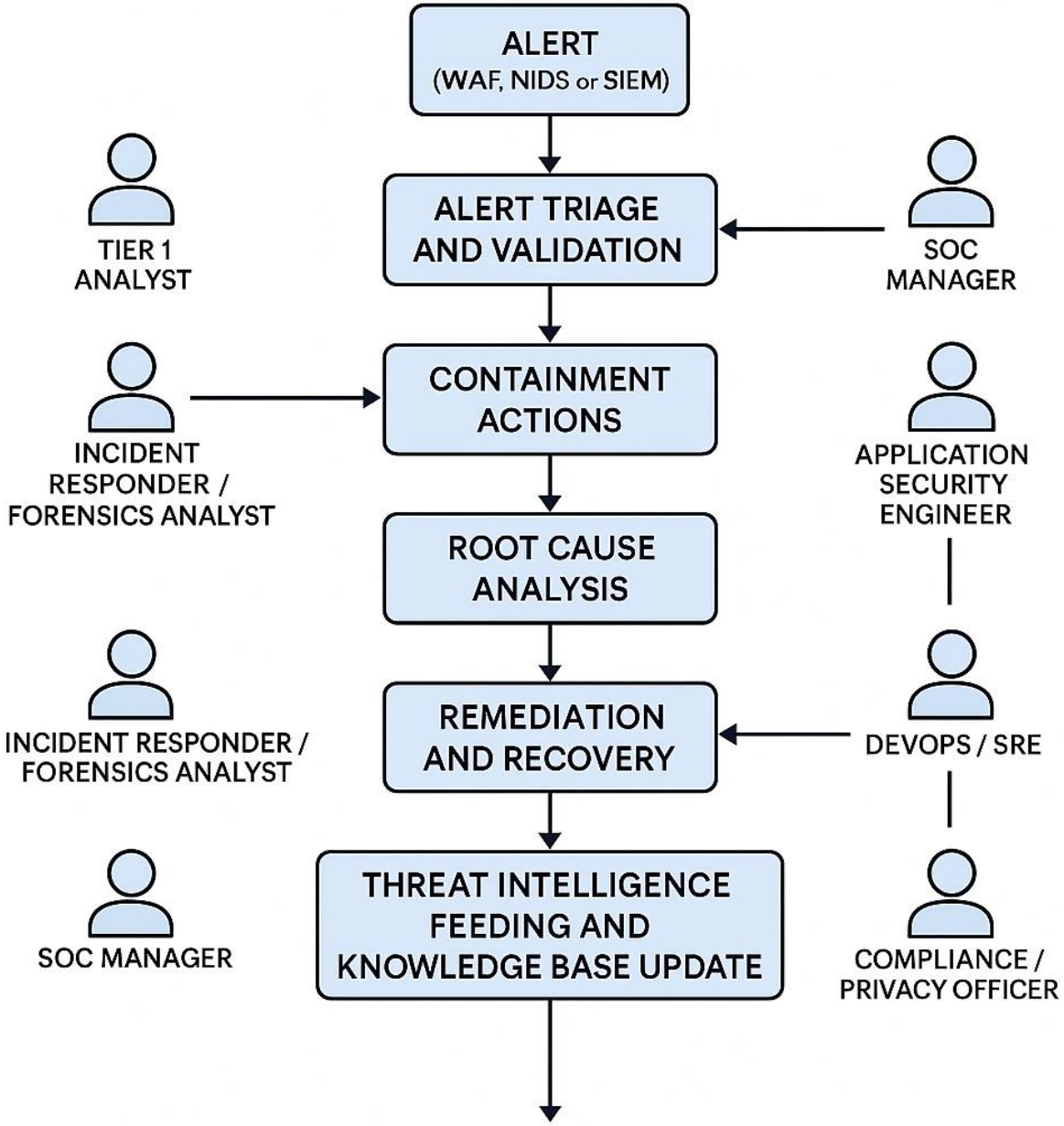
Employing RAG models to access up-to-date cybersecurity knowledge bases or internal documentation, providing timely, accurate, and contextually relevant answers during security incidents or routine operations.

Threat intelligence application is the target for first phase, with incident response, compliance automation, analysts support and RAG integration (1, 2, 5, 7 and 10) as potential targets thereafter. Models must be trained on Cyber security data (public + private) and then plugged into the environment for proper contextualization and customization of output

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We have potential threats originating from multiple systems, each of which requires its own identity management system (potentially integrated but not so simple), its own interface and analytics, and each of which requires in depth understanding of the system in question and the specific domain being protected.

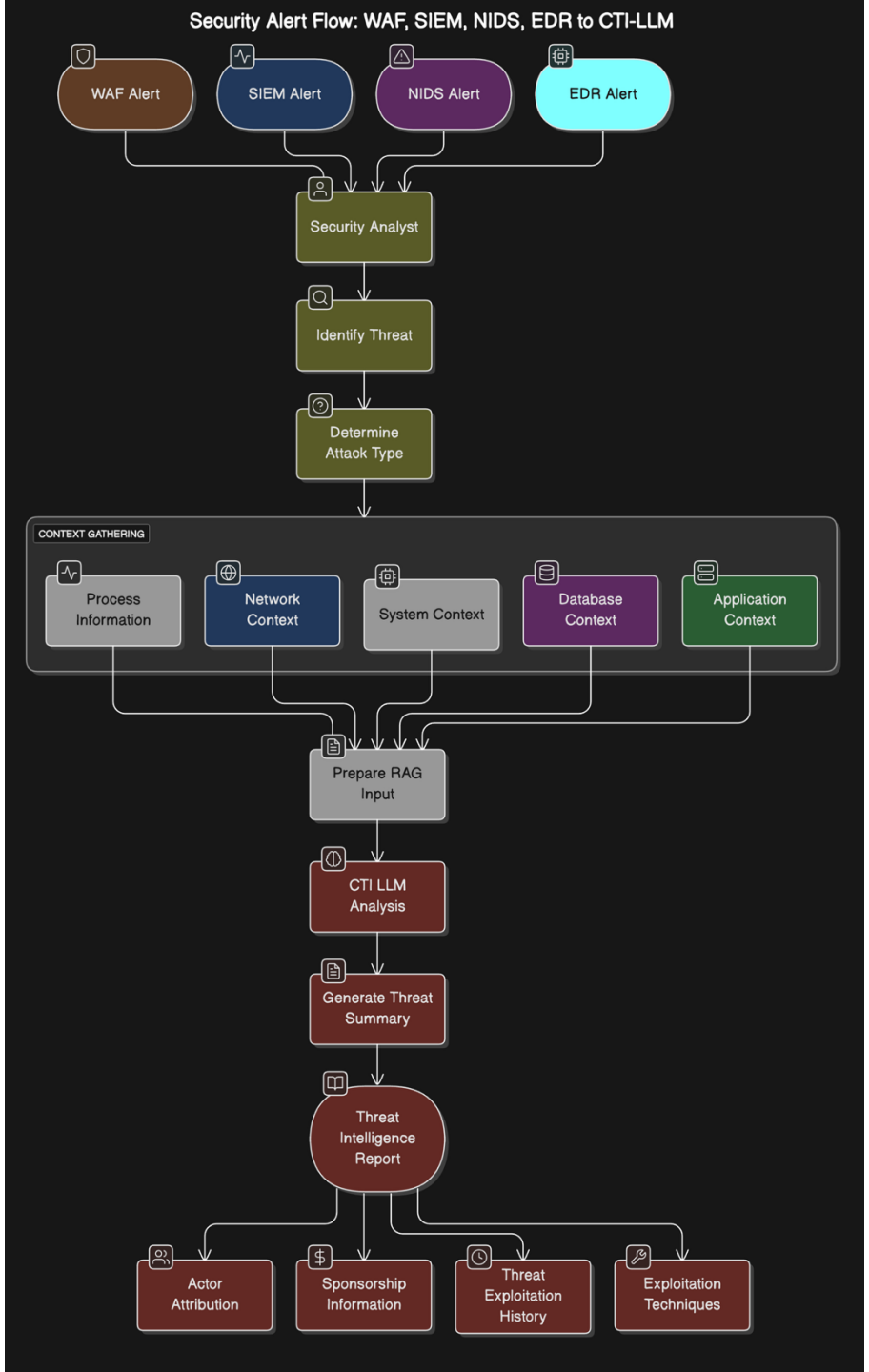


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Stop the madness!

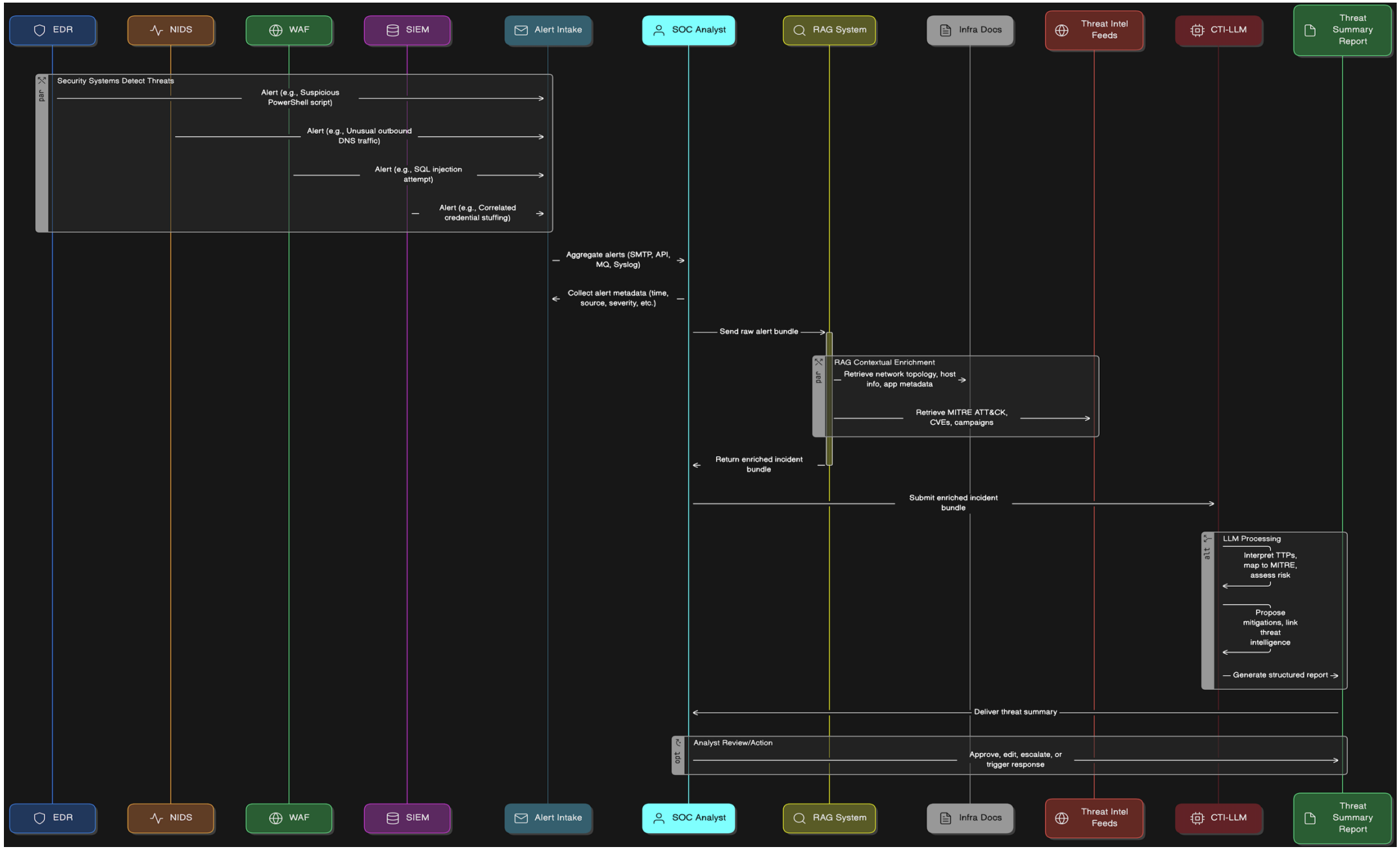
Consolidate threat sources and normalize the data. Feed into CTI-LLM that collates the detailed technical info with best-in-class industry knowledge (Ignite).

Enter Athena…



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End to End Threat Intake and Response with Athena's CTI-LLM



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Cyber Security Language Models

Model Name	Introduced	Developer(s)	Base Model	Training Methodology	Targeted Domains
CyberPal.AI	Aug-24	Matan Levi et al.	Not specified	Instruction tuning with SecKnowledge dataset; evaluated using SecKnowledge-Eval benchmark	Threat hunting, malware analysis, vulnerability assessment, incident response
Llama-Primus	Feb-25	Trend Micro AI Lab	Llama-3.1-8B-Instruct	Continual pretraining on cybersecurity corpus; instruction fine-tuning with curated QA tasks	Cyber threat intelligence, vulnerability classification, security certification preparation
SecBERT	2022	jackaduma	BERT	Pretrained from scratch on cybersecurity texts including APTnotes, Stucco-Data, CASIE, and SemEval-2018 Task 8; custom vocabulary	Threat intelligence, NER, text classification, semantic understanding, Q&A in cybersecurity context
SecureBERT	2022	Ehsan Aghaei et al. (UNC Charlotte)	BERT	Continual pretraining with customized tokenizer on large corpus of cybersecurity text	Cyber threat intelligence, cyber automation, phishing detection, code and malware analysis, intrusion detection
CTI-BERT	Dec-23	IBM T.J. Watson Research Center	BERT	Pretrained from scratch on high-quality cybersecurity corpus containing CTI reports and publications	Cyber threat intelligence extraction, security entity (IoCs) extraction, attack technique (TTPs) classification
SecurityLLM	Jan-25	ZySec-AI	Zephyr-7b-beta	Direct Preference Optimization (DPO) training across 30+ security domains	Threat analysis, compliance verification, security architecture reviews, SOC operations, risk management
SecurityBERT	Jun-23	Mohamed Amine Ferrag et al.	BERT	Privacy-preserving encoding technique (PPFLE) combined with Byte-level Byte-Pair Encoder (BBPE) Tokenizer; trained on IoT data	Cyber threat detection in IoT/IIoT devices

Variety of models have been introduced with a broad range of test methodologies and applications. Technology development and adoption is still in its infancy.

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Evaluating the Performance of Cyber Security LLMs

Benchmarks for Cyber Security Language Models

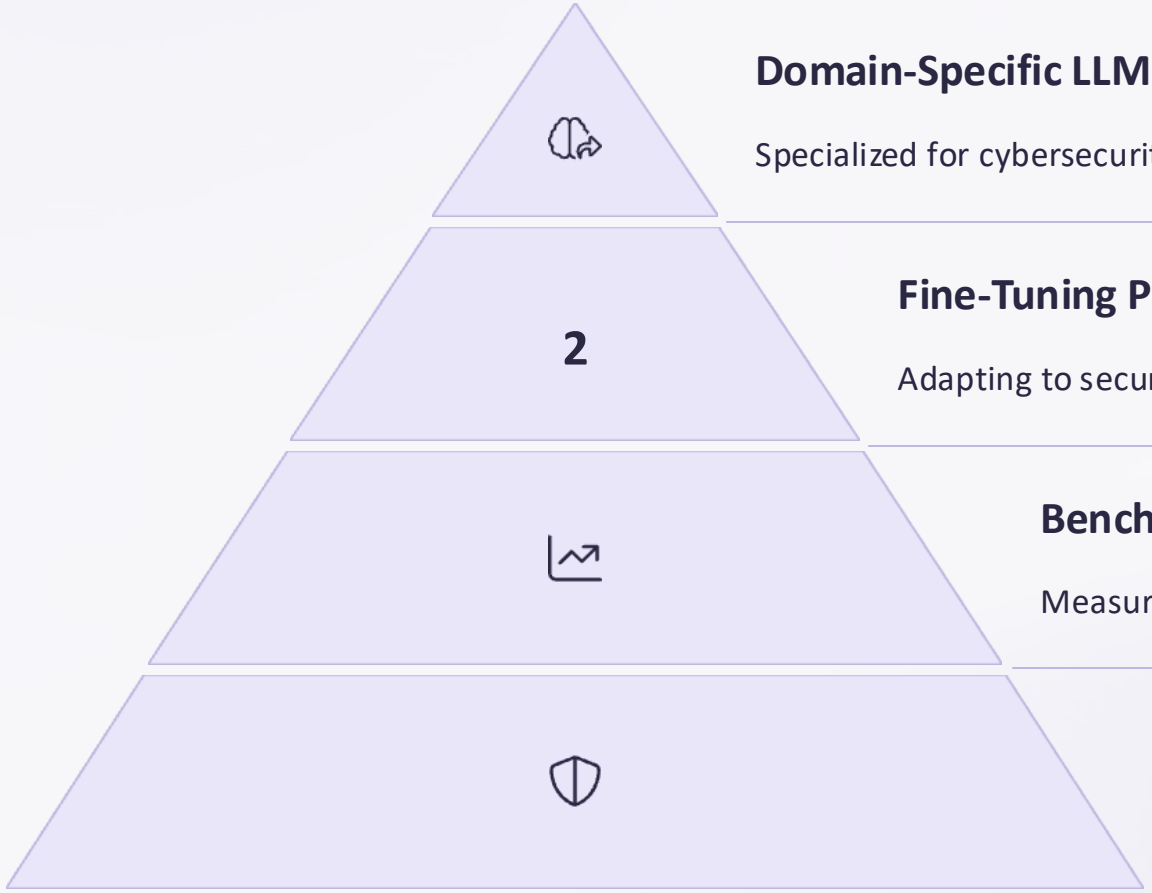
Benchmark	Organization / Authors	Open Source	Cybersecurity Domain	Key Tasks	Introduced In
CTIBench	University of Waterloo & AI4Sec	Yes (GitHub, Hugging Face)	Threat Intelligence, Vulnerability Analysis, Threat Attribution	CTI-MCQ, CTI-RCM, CTI-VSP, CTI-TAA	Jul-24
SecKnowledge-Eval	CyberPal.AI (Authors of SecKnowledge Paper)	No (Contact authors)	Instruction Following, Threat Analysis, Security Reasoning	Threat/attack pattern understanding, contextual security tasks	Aug-24
CyberSecEval 2	Meta AI	No	Offensive & Defensive Cybersecurity, LLM Risk Assessment	Malicious instruction handling, threat detection, security training evaluation	Apr-24
Sophos Benchmark Suite	Sophos AI	No	SOC Operations, Incident Handling	Incident summarization, SQL generation, severity rating	Mar-24
SECURE	Rochester Institute of Technology	Partially (some datasets)	ICS Security, Threat Reasoning	Knowledge extraction, understanding, reasoning in ICS scenarios	May-24
Cybench	CTF Community Researchers	Limited (some CTF challenges)	Offensive Security, Exploit Development	CTF task solving, vulnerability discovery, exploit execution	Jun-24

Evaluation of open-source benchmarks, applicability to threat intelligence specifically and potentially expanding and creating our own benchmark that speaks to the capabilities around Flashpoint Threat Intel specifically.

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Base-model Selection & Training



Domain-Specific LLM

Specialized for cybersecurity analysis

Fine-Tuning Pipeline

Adapting to security tasks

Benchmarking Framework

Measuring security-specific performance

Safety Guardrails

Ensuring responsible AI use

Leveraging Large Language Models (LLMs) for tasks like summarizing threat intelligence, explaining alerts, or answering analyst questions requires selecting a suitable base model and fine-tuning it on domain-specific data.

Benchmarking Base Models

CTI-Bench Task	Zephyr-7B-beta (F1/Acc)	Llama-3-8B-Instruct (F1/Acc)	Sec-Gemini-v1 (F1/Acc)	Metric Type
IOC Extraction	0.75	0.80	0.82	F1
TTP Identification	0.68	0.72	0.75	F1
Report Summarization	0.40	0.45	0.48	ROUGE-L
Vulnerability Mapping	0.85	0.88	0.90	Accuracy
Overall Rank	3	2	1	-

Before investing in fine-tuning, evaluate candidate base models out-of-the-box on relevant benchmarks. Proposed models: Zephyr-7B-beta, Llama-3-8B-Instruct, Sec-Gemini-v1 (assuming availability).

CTI-Bench is suggested, likely containing tasks relevant to Cyber Threat Intelligence (CTI) use cases (e.g., IOC extraction, TTP identification, threat report summarization).

Based on illustrative results, Sec-Gemini-v1 shows the best OOTB performance, followed by Llama-3-8B. This quantitative data provides a strong basis for selecting the most promising base model for subsequent fine-tuning.

Example: CyBERT (2021)

Contextualized Embeddings for the Cybersecurity Domain

The primary contribution of this paper is providing the security community with an initial fine-tuned BERT model that can perform a variety of cybersecurity-specific downstream tasks with high accuracy and efficient use of resources.

We create a cybersecurity corpus from open-source unstructured and semi-unstructured Cyber Threat Intelligence (CTI) data and use it to fine-tune a base BERT model with Masked Language Modeling (MLM) to recognize specialized cybersecurity entities.

We evaluate the model using various downstream tasks that can benefit modern Security Operations Centers (SOCs).

The fine-tuned CyBERT model outperforms the base BERT model in the domain-specific MLM evaluation. We also provide use-cases of CyBERT application in cybersecurity based downstream tasks.

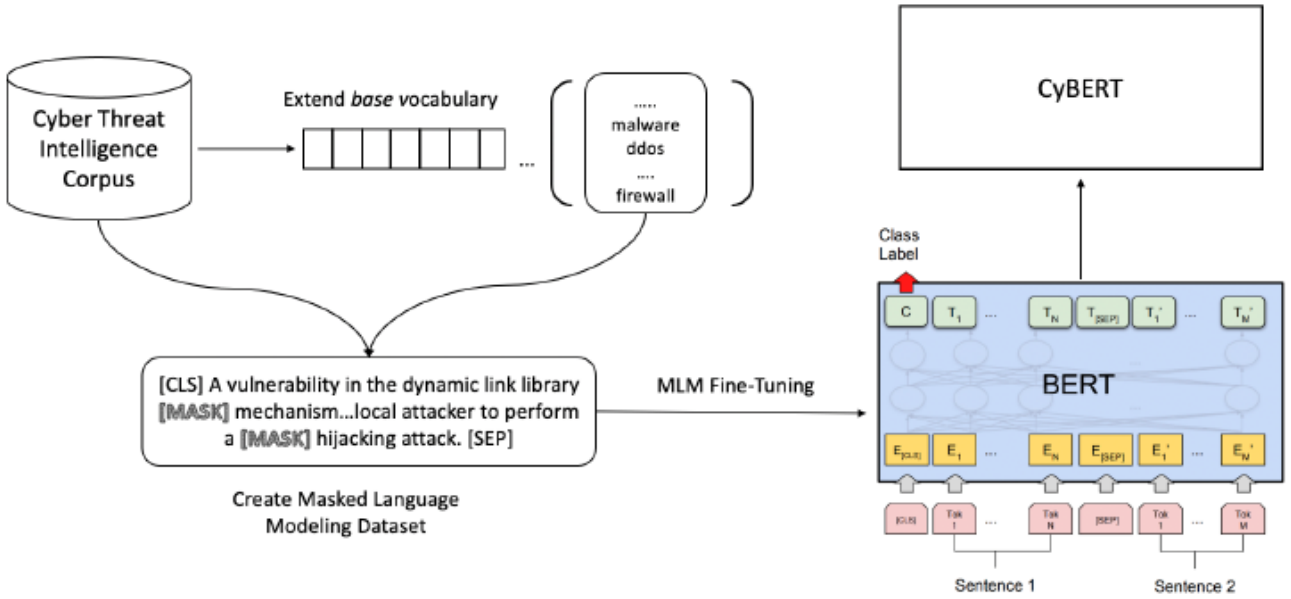


Fig. 1. Architecture Diagram for fine-tuning of the pre-trained BERT model to create CyBERT.

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P. Ranade, A. Piplai, A. Joshi and T. Finin, "CyBERT: Contextualized Embeddings for the Cybersecurity Domain," *2021 IEEE International Conference on Big Data (Big Data)*, Orlando, FL, USA, 2021, pp. 3334-3342, doi: 10.1109/BigData52589.2021.9671824.

LLM Training Optimization Techniques

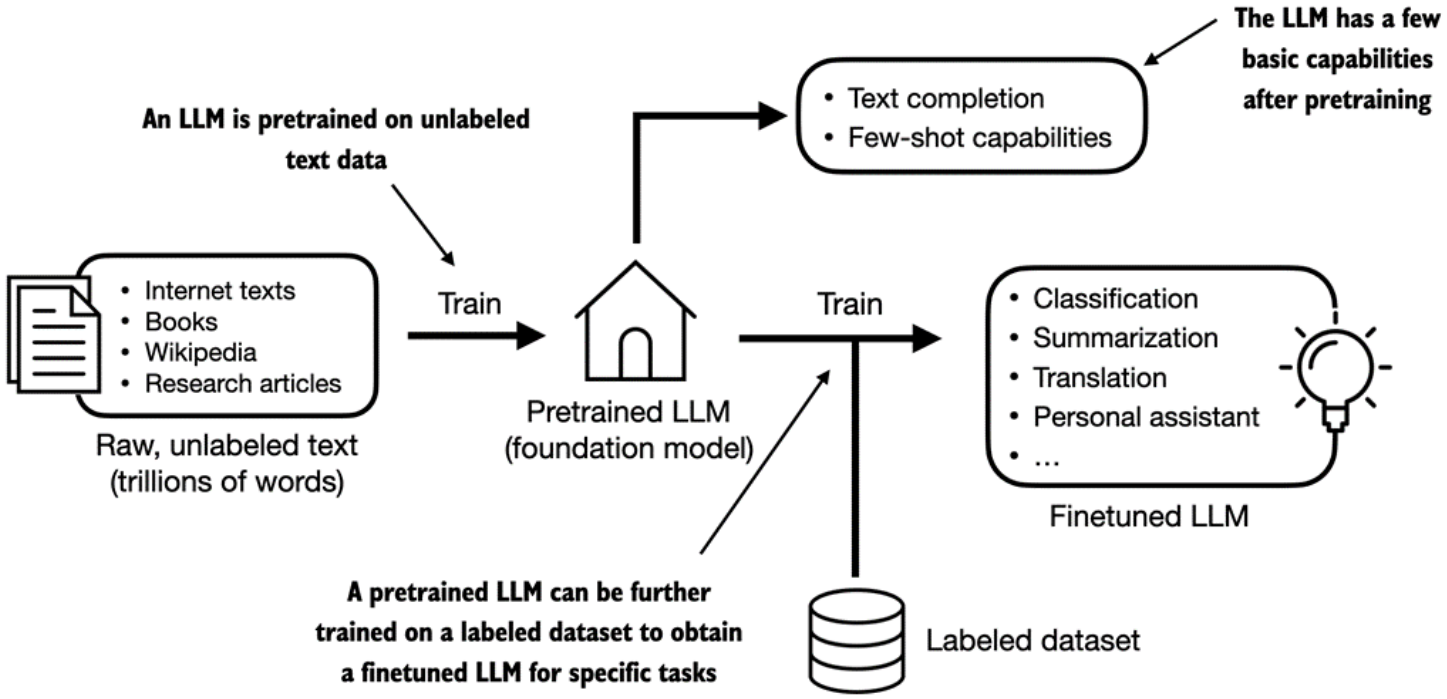
APPROACHES TO OPTIMIZING LLM MODEL SIZE & COMPUTE CHARACTERISTICS

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Pretraining vs Finetuning

The general process of creating an LLM includes pretraining and finetuning. The term "pre" in "pretraining" refers to the initial phase where a model like an LLM is trained on a large, diverse dataset to develop a broad understanding of language

This pretrained model then serves as a foundational resource that can be further refined through finetuning, a process where the model is specifically trained on a narrower dataset that is more specific to particular tasks or domains.



PEFT: Parameter Efficient Fine Tuning

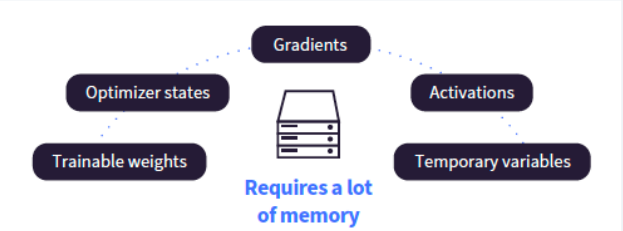
- PEFT stands for Parameter Efficient Fine-tuning.
- ML models are essentially complex mathematical equations with numerous coefficients or weights.
- These coefficients are responsible for the model behavior and make it capable of learning from data.
- During training of ML models, we adjust these coefficients to minimize errors and make accurate predictions.
- In case of LLMs, which can have billions of parameters, and changing all of them during training can be computationally expensive and memory-intensive.
- PEFT, as a subset of fine-tuning, takes the parameter efficiency seriously.
- Instead of altering all the coefficients of the model, PEFT selects a subset of them.
- It helps us significantly reducing the computational and memory requirements.

data iku

Parameter Efficient Fine-Tuning (PEFT) Methods

PEFT

Full fine-tuning of LLMs is challenging:



PEFT methods **only update a small number of model parameters.**
Examples of PEFT techniques:

- Freeze most model weights, and **fine tune only specific layer parameters.**
- Keep existing parameters untouched; **add only a few new ones or layers** for fine-tuning.

→ The trained parameters can account for only 15%-20% of the original LLM weights.

Main benefits:

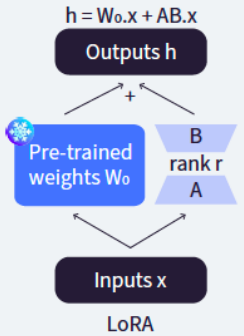
- Decrease memory usage, often requiring just 1 GPU.
- Mitigate risk of catastrophic forgetting.
- Limit storage to only the new PEFT weights.

Multiple methods exist with trade-offs on parameters or memory efficiency, training speed, model quality, and inference costs.
Three PEFT methods classes from literature:

Selective	Reparameterization	Additive
Fine-tune only specific parts of the original LLM.	Use low-rank representations to reduce the number of trainable parameters. *E.g., LoRA*	Augment the pre-trained model with new parameters or layers, training only the additions. → Adapter → Soft prompts

LoRA

Method to reduce the number of trainable parameters during fine-tuning **by freezing all original model parameters** and injecting a **pair of rank decomposition matrices** alongside the original weights



1 - Keep the majority of the original LLM weights frozen.
2 - Introduce a pair of **rank decomposition matrices.**
3 - Train the new matrices A and B.

Model weights update:
1 - Matrix multiplication:

$$B * A = BxA$$

2 - Add to original weights :

$$W_0 + BxA$$

Additional notes:

- No impact on inference latency.
- Fine-tuning specifically on the **self-attention layers** using LoRA is often enough to enhance performance for a given task.
- Weights can be switched out as needed, allowing for training on **many different tasks.**

Rank Choice for LoRA Matrices:

Trade-Off: A smaller rank reduces parameters and accelerates training **but** risks lower adaptation quality due to reduced task-specific information capture.

*In literature, it appears that a **rank between 4-32** is a good trade-off.*

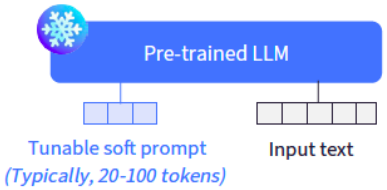
LoRA can be combined with quantization (=QLoRA).

SOFT PROMPTS

Unlike prompt engineering, whose limits are:

- The manual effort requirements
- The length of the context window

Prompt tuning: Add trainable tensors to the model input embeddings, commonly known as "soft prompts," optimized directly through gradient descent.



Soft prompt vectors:

- Equal in length to the embedding vectors of the input language tokens
- Can be seen as **virtual tokens** which can take any value within the multidimensional embedding space

In prompt tuning, LLM weights are frozen:

- Over time, the embedding vector of the soft prompt is adjusted to optimize model's completion of the prompt
- Only **few parameters are updated**
- A different set of soft prompts can be trained for each task and easily swapped out during inference (occupying very little space on disk).

From literature, it is shown that at 10B parameters, prompt tuning is as efficient as full fine-tuning.

⚠ *Interpreting virtual tokens can pose challenges (nearest neighbor tokens to the soft prompt location can be used).*

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LoRA & QLoRA PEFT Methods

LoRA (Low-Rank Adaptation)

Injects small, trainable low-rank matrices into the model's layers. Only these small matrices are updated, drastically reducing the number of trainable parameters.

The rank parameter (r) controls the size (rank) of the LoRA matrices. Common values are 8, 16, 32, 64. Higher ranks allow capturing more complex adaptations but increase trainable parameters and memory.

QLoRA

Combines LoRA with quantization (e.g., 4-bit NormalFloat) to further reduce memory footprint during training, allowing fine-tuning of larger models on less hardware.

Fine-tuning typically involves applying LoRA/QLoRA to specific layers (often attention layers) with a chosen rank and alpha parameter (scaling factor).

Parameter-Efficient Fine-Tuning (PEFT) methods allow adapting large models with significantly less computational cost than full fine-tuning.

Using QLoRA with a moderate rank (e.g., 16) is a common and effective strategy for balancing adaptation quality with resource efficiency when efficiency when fine-tuning models in the 7B-13B parameter range. Experimentation is needed to find the optimal rank and layers to target. layers to target.

LoRA and QLoRA for Coefficient Selection

LoRA (Low-Rank Adoption):

It is a technique exploits the fact that some weights have more significant impacts than others. In LoRA, the large weight matrix is divided into two smaller matrices by factorization.

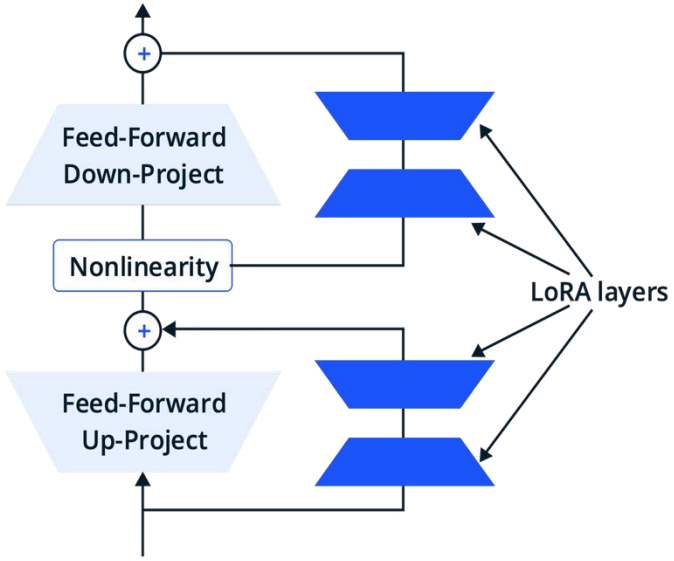
We reduce the number of coefficients that need adjustment, making the fine-tuning process more efficient.

QLoRA (Quantization + Low-Rank Adoption):

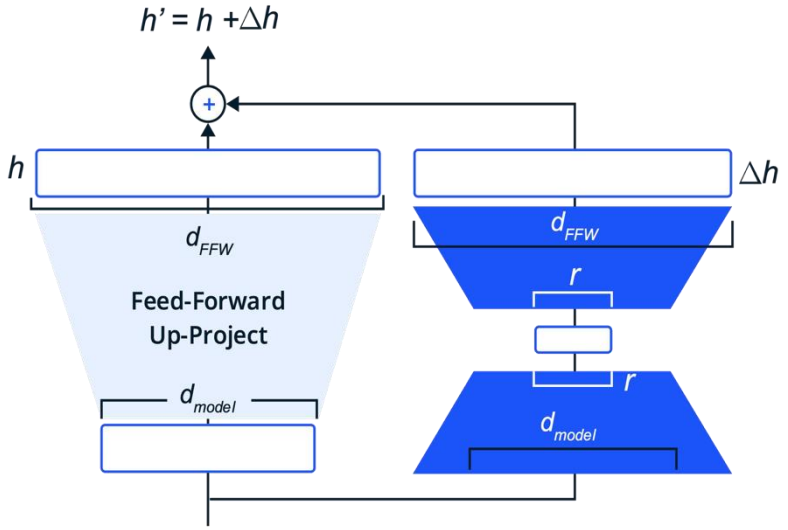
Quantization involves converting high-precision floating-point coefficients into lower-precision representations, such as 4-bit integers.

Quantization offers a solution by reducing the precision of these coefficients.

For instance, a 32-bit floating-point number can be represented as a 4-bit integer within a specific range. This conversion significantly shrinks the memory footprint.



LeewayHertz



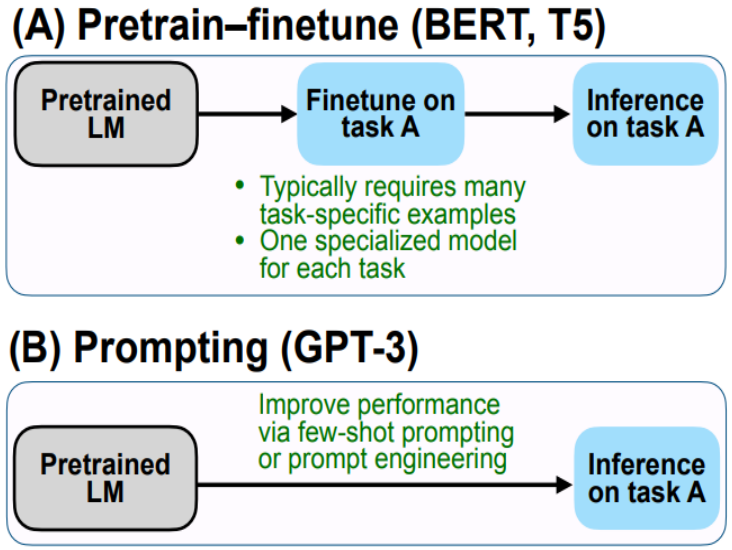
LeewayHertz

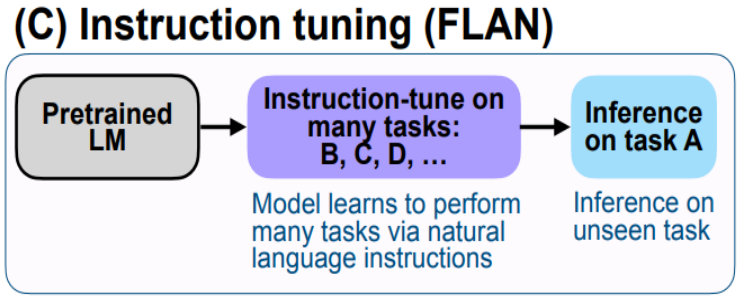
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Instruction Tuning

Finetuned Language Net (FLAN)

- Whereas supervised fine-tuning trains models on input examples and their corresponding outputs, instruction tuning augments input-output examples with instructions, which enables instruction-tuned models to generalize more easily to new tasks.
- "Instruction tuning", which finetunes language models on a collection of tasks described via instructions, substantially boosts zero-shot performance on unseen tasks.
- Methods differ based on how the instruction tuning data is constructed.

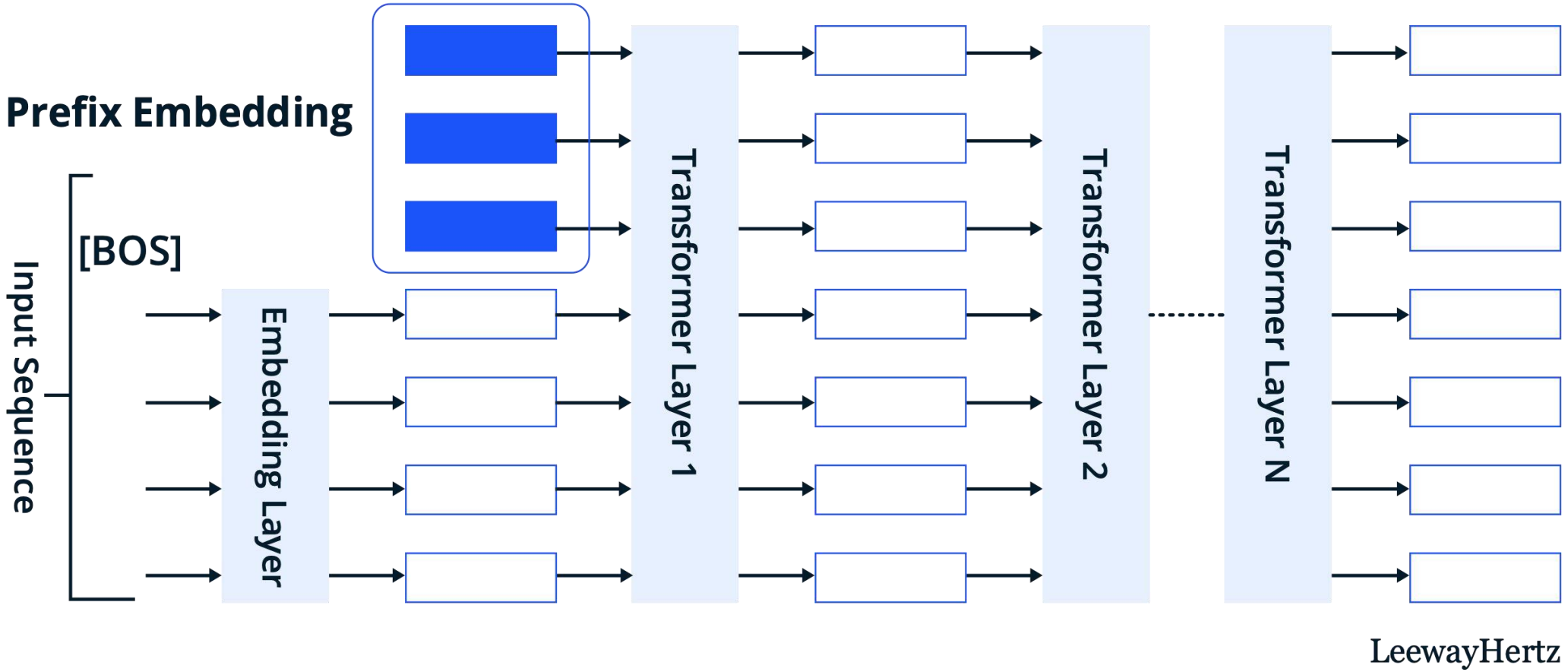




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Finetuned Language Models Are Zero-Shot Learners; Wei et al. 2022

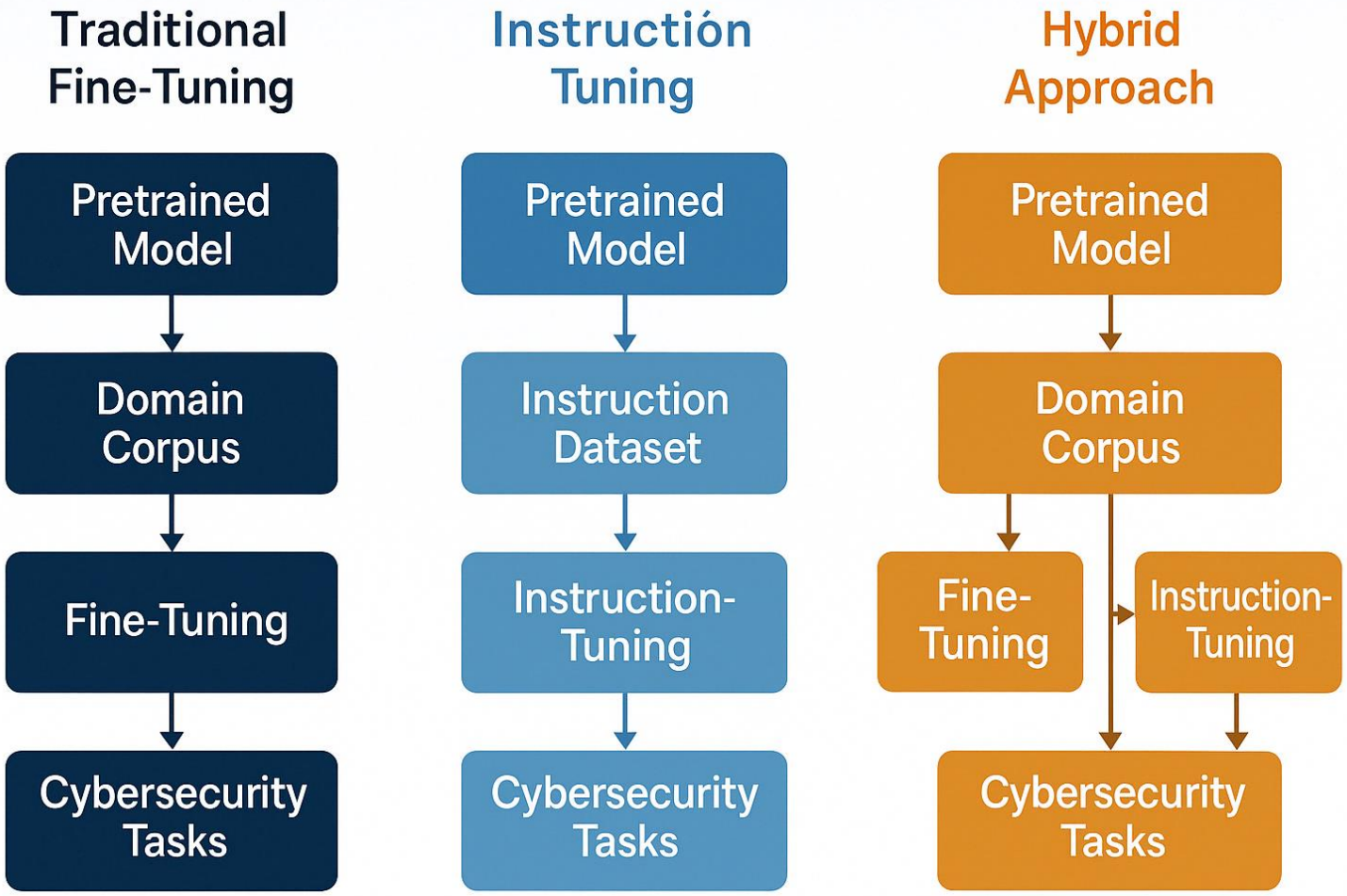
Prompt Tuning



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https://www.leewayhertz.com/parameter-efficient-fine-tuning/

Tuning Approaches



Different Training Strategies

Aspect	Traditional Fine-Tuning	Instruction Tuning	Hybrid Approach
Description	Adapting a general-purpose model to cybersecurity by updating weights on a large unlabeled/labeled **domain corpus**	Adapting a model to follow **explicit instructions** in natural language format across various cybersecurity tasks	Combines traditional fine-tuning on a domain corpus followed by instruction tuning or integrates both into a single multi-objective training setup
Input Format	Raw cybersecurity texts (blogs, CVEs, advisories)	Tuples like *("Map this CVE to MITRE ATT&CK", CVE text → expected answer)*	First: domain texts → corpus adaptation; then: curated instructions → multi-task generalization
Training Objective	Masked Language Modeling (MLM) or Causal Language Modeling (CLM); optionally with classification/regression heads	Supervised generation / classification using cross-entropy loss	Pretrain or adapt with MLM, then instruction tune; or mix the objectives (e.g., multi-task loss)
Best For	Embedding learning, classification, entity recognition, internalizing domain-specific syntax and jargon	Multi-turn reasoning, question-answering, structured task execution (e.g., IOC extraction, root cause summarization)	Use cases needing both strong domain understanding and complex task following (e.g., SOC assistant, security triage)
Sample Cyber LLMs	• CTI-BERT (IBM) • SecureBERT (UNC) • SecBERT (jackaduma) • SecurityBERT (IoT)	• CyberPal.AI • Llama-Primus • SecurityLLM • Sec-Gemini v1 (RAG + instruction)	• Llama-Primus (base → merged) • CTI-BERT + instruction adapters • CyberPal.AI w/ SecKnowledge → SecKnowledge-Eval loop
Strengths	✓ Captures nuanced cybersecurity terminology ✓ Improves base model for domain-specific NLP tasks	✓ Generalizes across task types ✓ Interpretable and human-aligned ✓ Easier downstream zero-shot/few-shot use	✓ High performance in both static understanding and reasoning ✓ Flexibility across tasks ✓ Lower label dependency if pretrained well
Weaknesses	✗ May not generalize well to unfamiliar task formats ✗ Doesn't teach "task framing"	✗ Requires high-quality curated instruction data ✗ May hallucinate if not grounded	✗ Expensive and complex training pipeline ✗ Risk of forgetting if improperly scheduled
Compute Requirements	• Requires **moderate GPU resources** for continual pretraining (e.g., 8x A100 for 100M tokens) • Good fit for distributed training	• Typically **less data but more curated**, though larger sequence lengths and decoder-based models (e.g., LLaMA, GPT) may demand **more memory** • 8x A100 for ~20K examples	• Highest resource requirement: pretraining on corpus (TPU/8x A100) + instruction finetuning (decoder model, RAG-enabled, ~24GB GPU RAM or more)
Parameter Efficient Tuning	Supported via LoRA, Adapters (sparse updates)	Works well with PEFT (e.g., QLoRA, LoRA), especially for instruction-tuning-friendly architectures	Common hybrid practice: LoRA for instruction head + frozen backbone or adapter fusion
Instruction-Awareness	✗ Model is unaware of instructions explicitly	✅ Instruction-following enabled	✅ Instruction-aware + domain-knowledge enriched
Fine-Tuning Duration	Long (10–40 hours for 100M tokens depending on model size)	Medium (4–12 hours for 20K–100K examples depending on format)	Longest (20–80 hours if running pretrain + instruction + eval loop with RAG)

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Adapting LLMs for Cyber Security

GradientFlow.com

Getting The Most Out Of Large Language Models: Tune, Prompt, Reward

	What	How	When
Fine-tuning	Entails taking a pre-trained language model and further training it on a specific & smaller dataset that is specific to the task at hand. This is typically done by updating the weights of the model's last layer or layers while leaving the rest of the model static.	During fine-tuning, a pre-trained model is loaded into memory and its weights are frozen. A smaller dataset relevant to the task at hand is loaded, and the pre-trained model is adjusted by tuning its weights. The model is typically trained for several epochs until the desired level of accuracy is reached.	The fine-tuning process is normally used when the task or domain is well-defined, and there is sufficient labeled data available to train on. If you have a large dataset and a specific task in mind, fine-tuning a language model is likely to be the most effective approach.
Prompt Engineering	Involves designing natural language prompts or instructions that can guide a language model to perform a specific task.	Select & arrange the words in a prompt or query to elicit a specific response from the model. Top-notch prompt engineers conduct experiments, systematically record their findings, and refine their prompts to identify essential components.	Best suited for tasks requiring a high level of precision and well-defined outputs. Prompt engineering can be used to craft a query that elicits a desired output. In some cases, prompt engineering can be used to improve the performance of a fine-tuned model by providing more guidance to the model during inference.
RLHF Reinforcement Learning from Human Feedback	Reinforcement Learning from Human Feedback (RLHF) involves training a model by receiving feedback from human evaluators.	The model is first trained on a dataset to establish a baseline level of performance. The model then generates a response to a prompt or query that is evaluated by a human. Feedback from the human evaluator is utilized to update the model's weights so that it can generate more accurate responses in the future.	RLHF is ideally suited when the task requires a high level of accuracy and the model needs to be trained on a wide variety of inputs. RLHF is particularly useful when there is very limited data that can be used to train the model, since the model can be trained on a wide range of inputs through human feedback.

Hybrid approaches are proving to be the most effective and involve a combination of traditional finetuning strategies along with the instruction based tuning and RLHF. Source: https://gradientflow.com/maximizing-the-potential-of-large-language-models/

Instruction Dataset

Manual Creation

Human experts (SOC analysts, CTI analysts) analysts) write high-quality prompts and and responses. Slow and expensive but but yields the best quality.

Weak Supervision

Use heuristics, templates, or existing structured data (e.g., alert fields + analyst notes) to automatically generate instruction pairs. Faster but lower quality, requires careful filtering.

Synthetic Self-Play

Use a powerful LLM (e.g., GPT-4) to generate instruction prompts and corresponding responses based on provided documents or guidelines. Can scale effectively but requires careful validation to ensure quality and avoid reinforcing biases.

Supervised fine-tuning (SFT) requires an instruction dataset: pairs of (prompt, desired_response) demonstrating how the model should behave for specific behave for specific tasks. Estimates range from 20k to 100k high-quality instruction examples. Quality often matters more than quantity. quantity.

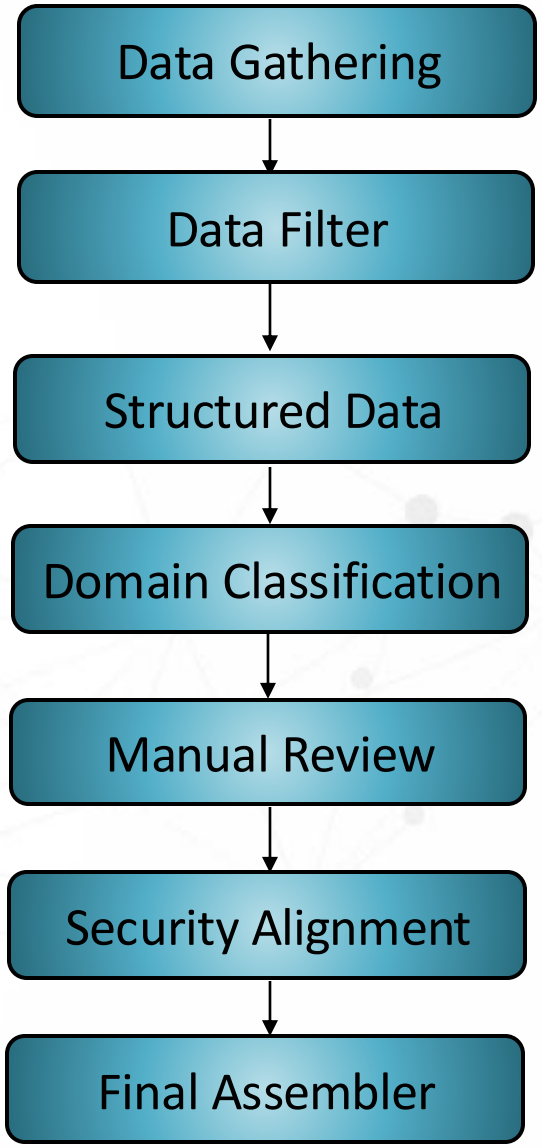
A combination of methods is often practical: start with a smaller set of high-quality, manually created examples covering core use cases, then augment cases, then augment with carefully validated synthetic data or weakly supervised examples to increase diversity and scale.

The focus should be on creating instructions that reflect the target tasks (e.g., summarizing an alert with environmental context, explaining an ATT&CK explaining an ATT&CK technique, correlating IOCs).

Data Gathering & Preprocessing Pipeline

Following are the steps that are included in data gathering and preprocessing pipeline:

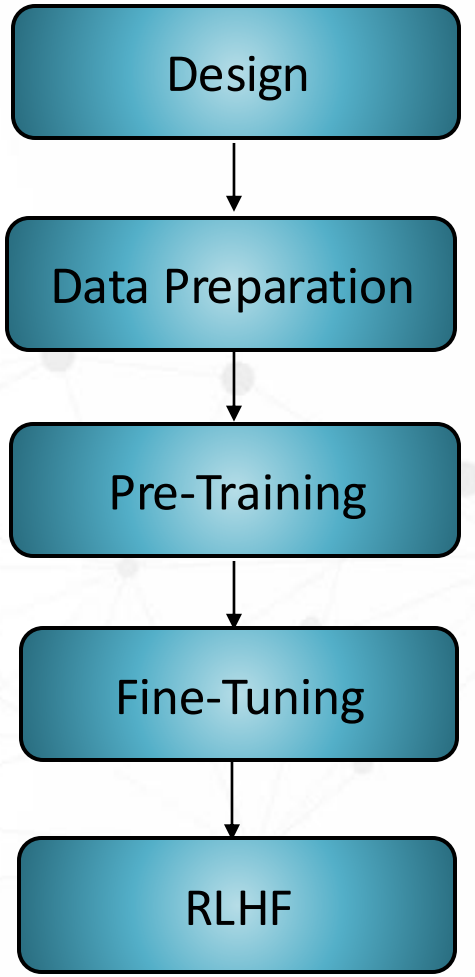
1. **Data Gathering**: Collect data from Cybersecurity websites.
2. **Data Filter**: Filters and only keeps useful data related to CTI.
3. **Structured Data**: Generate Question-Answer pairs from filtered data.
4. **Domain Classification**: Classifies data in relevant cybersecurity categories.
5. **Manual Review**: Human review of Instruction-response pairs.
6. **Security Alignment**: Enhance dataset with real scenarios.
7. **Final Assembler**: Merges final data, removes duplicates.



LLM Fine Tuning

Following are the steps that are included in LLM Fine Tuning:

1. **Design**: Architecture, Objectives, and Scope for Athena-CTI-LLM.
2. **Data Preparation**: Collect and prepare data as proposed in our data processing pipeline.
3. **Pre-training**: Train the base language model on a large corpus focused on CTI to learn domain-specific patterns.
4. **Fine-Tuning**: Use Parameter-Efficient Fine-Tuning (PEFT) to adapt the model for specific CTI tasks with Instruction-Response data.
5. **RLHF** (Optional): Apply reinforcement learning with human feedback to refine model responses for better alignment with expert expectations.



Threat Intelligence Data Integration

INTEGRATING FLASHPOINT IGNITE & OTHER DATA SOURCES INTO CUSTOM TRAINED LLM

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Cyber Security Use Cases leveraging Flashpoint Ignite

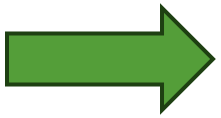
Actionable Threat Intelligence
- Real-time alerts on emerging cyber threats
- Indicators of Compromise (IOCs) and threat actor intelligence
- Curated insights into darknet and deep web activities

Risk Management and Vulnerability Assessment
- Detailed vulnerability information, CVE analysis, exploit availability
- Predictive insights to prioritize patching and security remediation

Cybercrime Monitoring
- Monitoring dark web forums, marketplaces, and illicit communities
- Early detection of credential leaks, fraudulent activities, and malicious campaigns

Integrations and API
- Seamless integrations with existing security stacks (SIEM, SOAR, endpoint security solutions)
- Rich API for threat feeds and automation workflows

Malware & Threat Actor Profiling
- Detailed threat actor reports, attribution, and behavior analysis
- Insights into geopolitical threats, hacktivism, organized crime, and nation-state actors
- Latest Malware exploits & Attack Vectors



Real-Time Threat Intelligence Analysis & Summarization
- Automatically generate concise summaries of new threats and intelligence reports.
- Reduces analyst time spent reviewing large threat intel documents.

Automated Incident Triage & Root Cause Analysis
- Objective: Leverage the LLM to swiftly analyze incidents by correlating alerts with known vulnerabilities, threat actors, and recent intelligence.
- Benefit: Accelerates response time and provides immediate contextual understanding.

Proactive Vulnerability & Exploit Prioritization
- Objective: Analyze CVEs, exploit availability, and real-world threat activity to prioritize patches and responses.
- Benefit: Improves security posture by aligning efforts with real-world risk.

Threat Actor Attribution & Behavioral Analysis
- Objective: Generate threat actor profiles from detailed data, recognizing patterns, tools, and tactics.
- Benefit: Enhances attribution accuracy and threat prediction capabilities.

Expected Benefits

✓ *Accelerate security operations by significantly reducing the time-to-analysis for threat intelligence.*

✓ *Enhance analytical accuracy through real-time integration with evolving threat data.*

✓ *Improve cybersecurity decision-making, prioritization, and proactive response.*

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Corpus Sizing

Flashpoint Data

Rich, real-world threat actor communications and CTI. Volume can be significant but requires careful filtering, sanitization, and adherence to licensing and compliance constraints.

Open Datasets

Public CTI feeds (e.g., MISP MISP galaxies, AlienVault AlienVault OTX), security security blogs, academic academic papers, vulnerability databases (NVD), malware analysis analysis reports (VirusTotal, sandbox reports), sanitized security security logs.

Sizing Target

The total corpus size should ideally be in the billions of tokens to effectively adapt the LLM. LLM. A corpus of 10B+ tokens would be a strong strong target.

Quality vs. Quantity Quantity

Quality and relevance are are as important as quantity. Focus on high-high-quality, domain-specific content.

Fine-tuning requires a substantial corpus of relevant text data. The proposal includes Flashpoint data (dark web, illicit communities - requires careful requires careful handling) and open datasets.

Quantifying the available tokens from Flashpoint (post-processing) and selected open datasets is necessary to determine if sufficient data exists or if further data sourcing is required.

Finetuning And Integration Strategy

Step 1: Domain Corpus Preparation

Data Sources; Flashpoint Ignite threat intelligence feeds (dark web forums, CVEs, threat actor reports, incident summaries). Structured indicators (IOC, TTPs, MITRE ATT&CK mappings).

Fine-Tuning Objective: Masked Language Modeling (MLM) to internalize threat intelligence vocabulary and cybersecurity-specific context.

Step 2: Curated Instruction Dataset Creation

Instructions Examples:

- "Summarize this threat report."
- "Attribute this incident to a known threat actor."
- "Identify potential impacts of this CVE."
- "Extract all relevant IOCs from this threat document."

Format: Structured as (Instruction, Input → Desired Output) tuples.

Step 3: Instruction Tuning

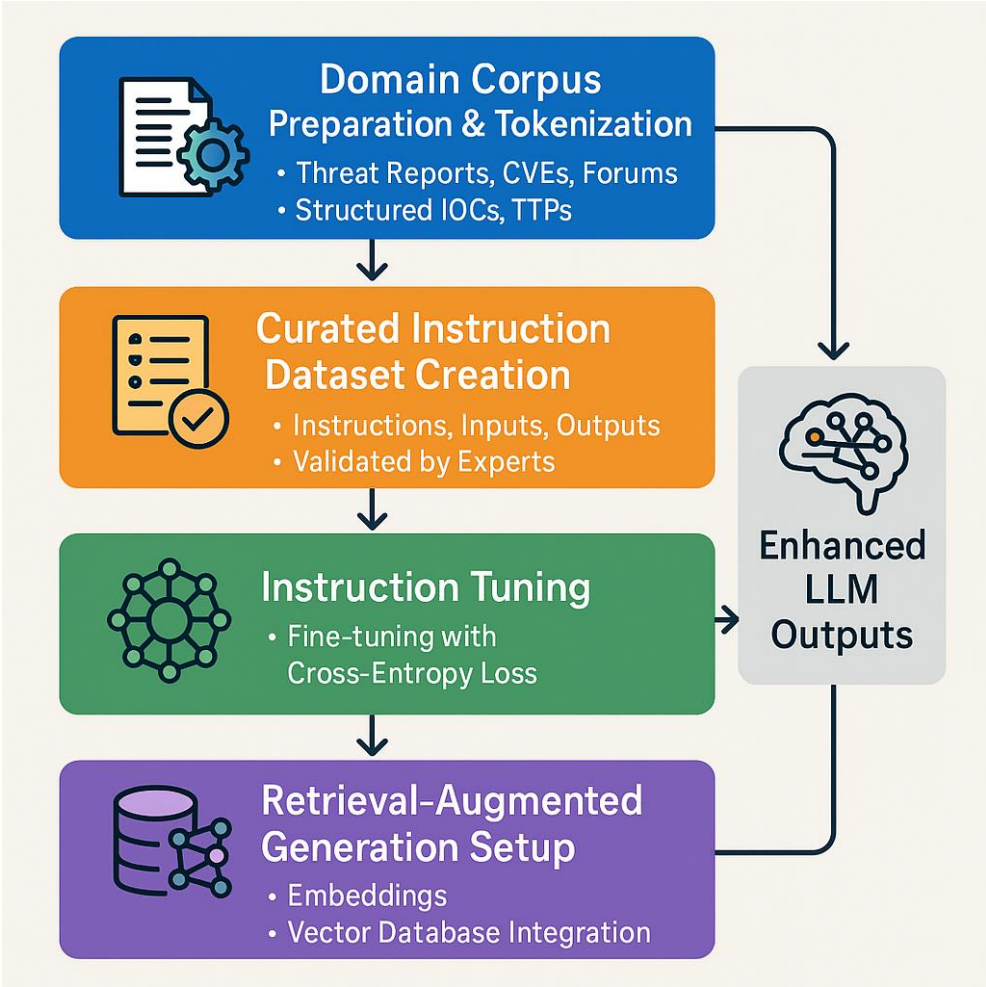
Objective: Enable the LLM to follow security-specific instructions.

Methodology: Supervised fine-tuning with instruction tuples, cross-entropy loss on generated outputs.

Step 4: Integration of Retrieval-Augmented Generation (RAG)

Combine the LLM with Ignite's real-time data retrieval via embeddings and vector stores.

Provide the model with external real-time context (latest threats, CVEs, etc.) to answer queries more accurately.



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LLM Model& Context Integration & Compute Requirements

Leverage state of the art Cyber Security LLMs and finetune against Flashpoint data using optimized finetuning strategies

Leverage state of the art Cyber Security LLMs with robust instruction-following capabilities and retrieval augmentation compatibility:

- Llama-Primus or Sec-Gemini v1 (instruction-tuned, cybersecurity-focused), or
- Alternative: Zephyr-7b-beta for efficiency.

PEFT: Utilize LoRA (Low-Rank Adaptation) or adapters for efficient updates and rapid deployment.

RAG Integration (Inference & Retrieval):

- Utilize vector stores (e.g., Neo-4j) for embedding-based real-time retrieval of environment context. (Minimal additional GPU overhead, primarily inference-focused.)

Compute Requirements (roughly)

Continual Pretraining (MLM):

- Moderate-to-large GPU resources (e.g., 8x NVIDIA A100 GPUs).
- Expected training duration: ~10–40 hours depending on corpus size (50M–200M tokens).

Instruction Tuning:

- 4–12 hours on 8x NVIDIA A100 GPUs for datasets of 20K–100K instruction tuples.

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Compute Budget Considerations (CONT)

$2-3

A100 GPU Cost

Per hour on cloud

100s

Training Hours

For full fine-tuning runs

4-8

Recommended GPUs

A100s/H100s for efficient training

50-70%

Potential Savings

Using spot instances

Fine-tuning LLMs, even with PEFT, requires significant GPU resources. Hardware options include on-premise GPUs (NVIDIA A100s or similar), cloud rented GPUs (NVIDIA H100s, GPUs (NVIDIA H100s, A100s), or specialized Cloud TPUs/Trainium.

Using cloud spot instances can significantly reduce cost but requires implementing robust checkpointing and restart strategies to handle potential interruptions.

For an 8B parameter model with QLoRA and a ~100k instruction dataset, fine-tuning might take several hours to days on a multi-GPU setup. The budget must accommodate initial must accommodate initial fine-tuning, hyper-parameter search, and ongoing weekly updates.

Sensitive Data Handling (Flashpoint, OFAC/ITAR)

Compliance

Be aware of potential OFAC (Office of Foreign Foreign Assets Control) or ITAR (International (International Traffic in Arms Regulations) implications if data involves sanctioned entities entities or export-controlled information. Consult legal counsel to ensure compliance compliance based on data specifics and user user locations.

Access Control

Implement strict, granular access controls (potentially leveraging graph DB labels or separate databases) to ensure only authorized personnel can access raw Flashpoint data or derivatives.

Encryption

Encrypt sensitive data both at rest (storage) and in transit. Consider application-level encryption for particularly sensitive fields.

Minimization

Process and store only the minimum data necessary for the intended intended analysis. Avoid broad collection if possible.

Sanitization/Anonymization

Implement procedures to sanitize or anonymize data before it's used for broader analysis or LLM training, where feasible and appropriate.

Ingesting sensitive third-party data like Flashpoint (dark web intelligence) requires careful handling due to content sensitivity and potential compliance obligations. compliance obligations.

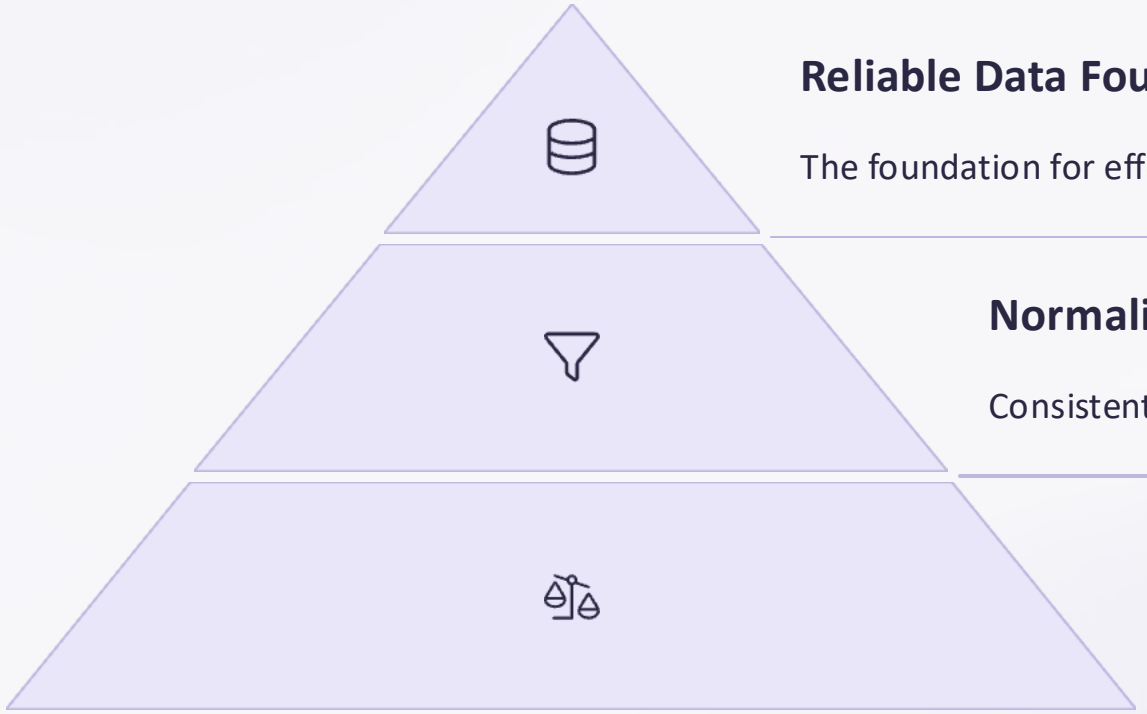
Handling data like Flashpoint requires robust technical controls and strict adherence to legal and contractual obligations.

LLM Product Lifecycle

METHODOLOGY: BUILD, EVALUATE, TEST, DEPLOY, ITERATE

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Data Ingestion Architecture Analysis



Reliable Data Foundation

The foundation for effective security analysis

Normalized Processing

Consistent handling of diverse data streams

Scalable Architecture

Designed to handle high volumes at enterprise scale

The foundation of an effective environmental-context aware cybersecurity platform lies in its ability to reliably ingest, normalize, and process normalize, and process diverse data streams at scale. This section evaluates key architectural choices for the data ingestion pipeline. pipeline.

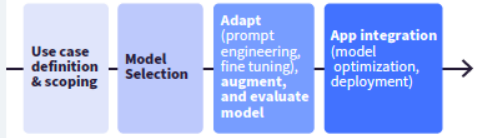
Finetuning strategy to be aligned with optimal compute requirements

data iku

LLM Compute Challenges and Scaling Laws

LARGE LANGUAGE MODEL CHOICE

Generative AI Project Lifecycle



Two options for model selection

- Use a pre-trained LLM.
- Train your own LLM from scratch.

But, in general...

...develop your application using a **pre-trained LLM,** except if you work with extremely specific data (i.e., medical, legal)

Hubs: Where you can browse existing models 🤗

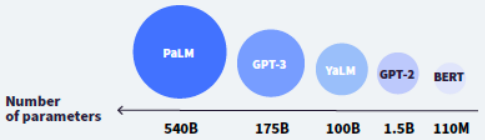
→ **Model Cards:** *List of best use cases, training details, limitations on models.*

The model **choice** will **depend on the details of the task** to carry out.

Model pre-training:

Model weights are adjusted in order to minimize the loss of the training objective.

It requires significant computational resources, (i.e., GPUs, due to high computational load).



COMPUTATIONAL CHALLENGES

Memory Challenge

```
RuntimeError : CUDA out of memory
```

→ LLMs are massive and require plenty of memory for training and inference.

To load the model into GPU RAM:

1 parameter (32-bit precision) = **4 bytes needed**
1B parameters = 4×10^9 bytes = 4GB of GPU

Pre-training requires storing additional components, beyond the model's parameters:

- Optimizer states (e.g., 2 for Adam)
- Gradients
- Forward activations
- Temporary variables

This could result in an additional **12-20 bytes of memory needed** per model parameter.

This would mean it requires **16 GB to 24 GB** of GPU memory to train a 1-billion parameter LLM, around **4-6x** the GPU RAM needed just for storing the model weights.

Hence, the memory needed for LLM training is:

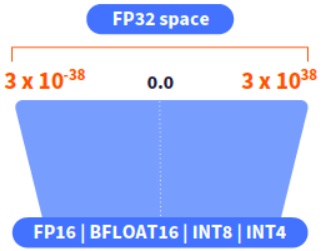
→ Excessive for consumer hardware

→ Even demanding for data center hardware (for single processor training).
For instance, NVIDIA A100 supports up to 80GB of RAM.

QUANTIZATION

How can you reduce memory for training?

Quantization: Decrease memory to store the weights of the model by converting the precision from 32bit to 16bit or 8bit integers.



Quantization maps the FP32 numbers to a lower precision space by employing scaling factors determined from the range of the FP32 numbers.

→ In most cases, quantization **strongly reduces memory requirements** with a **limited loss** in prediction.

BFLOAT16 is a popular alternative to FP16:

- Developed by Google Brain
- Balances memory efficiency and accuracy
- Wider dynamic range
- Optimized for storage and speed in ML tasks

e.g., FLAN T5 pre-trained using BFLOAT16

Benefits of quantization:

→ Less memory

→ Potentially better model performance

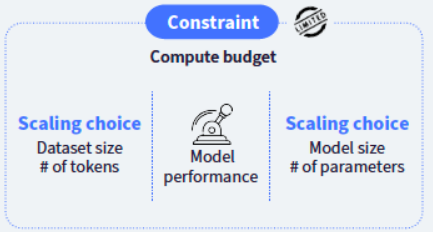
→ Higher calculation speed

SCALING LAWS

How big do the models need to be?
The goal is to maximize model performance.

Researchers explored trade-offs between the dataset size, the model size, and the compute budget:

Increasing compute may seem ideal for better performance, but practical constraints like hardware, time, and budget limit its feasibility.



It has been empirically shown that, as the compute budget remains fixed:

→ **Fixed model size:** Increasing training dataset size improves model performance.

→ **Fixed dataset size:** Larger models demonstrate lower test loss, indicating enhanced performance.

What's the optimal balance?

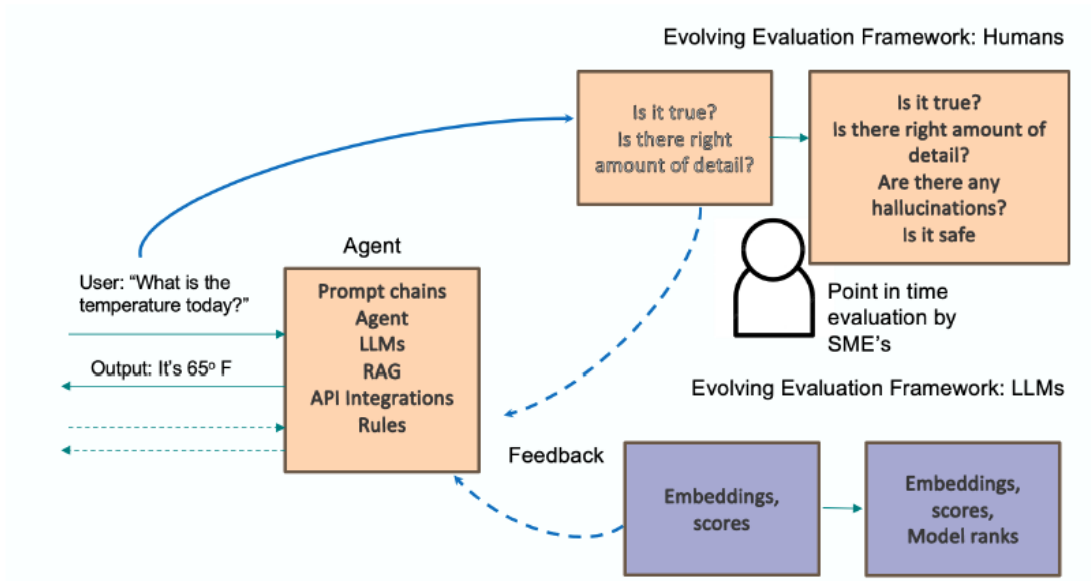
Once scaling laws have been estimated, we can use the **Chinchilla approach**, i.e., we can choose the dataset size and the model size to train a **compute-optimal model**, which maximizes performance for a given compute budget. The compute-optimal training dataset size is ~20x the number of parameters.

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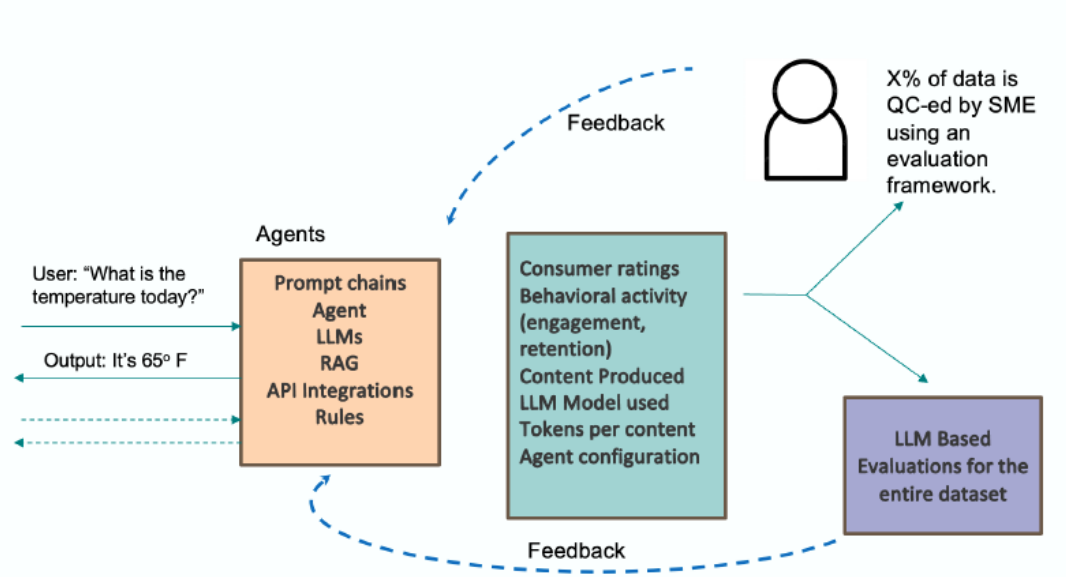
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Building a robust Evaluation Framework

Offline



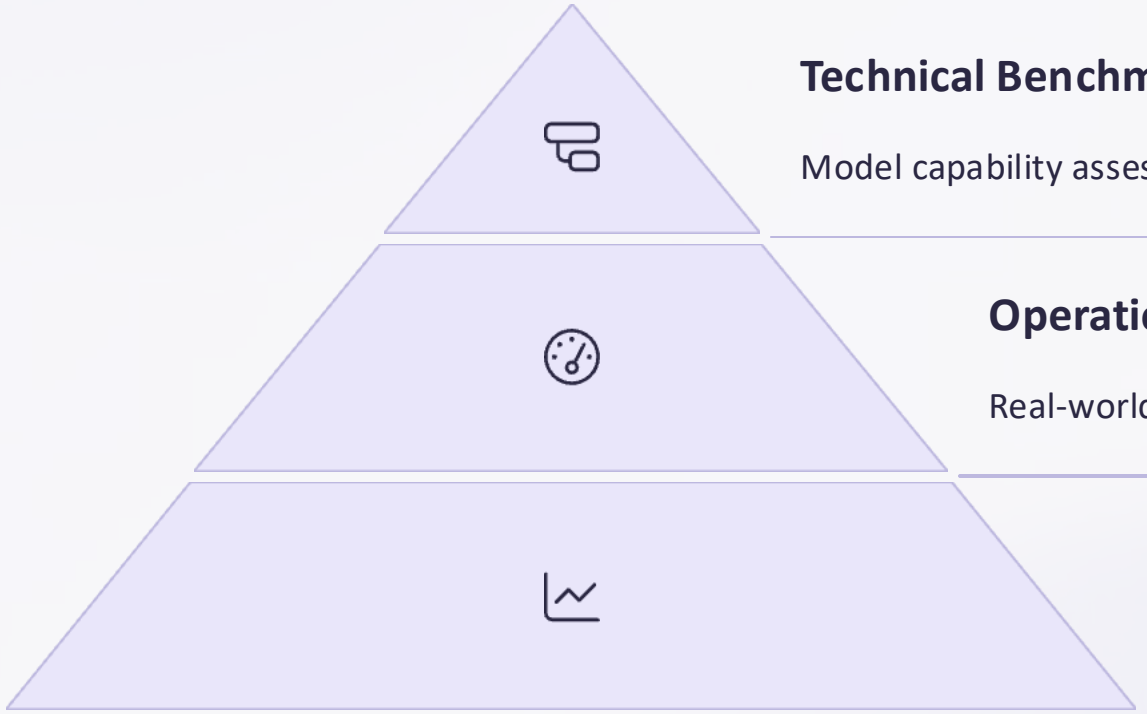
Online



Constantly evolves: needs versioning

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Benchmark & KPI Framework



Technical Benchmarks

Model capability assessment

Operational KPIs

Real-world performance metrics

Continuous Evaluation

Ongoing performance tracking

A robust framework for benchmarking model performance and measuring Key Performance Indicators (KPIs) is essential to evaluate success, evaluate success, guide improvements, and demonstrate value.

Mapping CTI-Bench Tasks to Use Cases

Use Cases

- Summarize incoming threat reports
- Extract IOCs from alerts/reports
- Identify ATT&CK techniques used
- Answer analyst questions about threats

CTI-Bench Tasks

- Report Summarization (ROUGE score)
- IOC Extraction (F1 score)
- TTP Identification (F1/Accuracy)
- Question Answering (Accuracy/F1)

The chosen benchmark suite (e.g., CTI-Bench) should contain tasks that directly map to the intended use cases of the LLM within the platform.

Define "good enough" performance thresholds (e.g., F1 > 0.80 for IOC Extraction, Accuracy > 0.75 for TTP Identification) based on operational requirements and baseline performance. These thresholds determine if a model version is suitable for deployment.

Ensure the benchmark covers the critical capabilities expected from the LLM in production. If existing benchmarks lack coverage for specific use cases, custom evaluation tasks may need to be developed.

Simulating Environment-Grounded Questions

Create Representative Data Data Snapshot

Prepare a realistic sample of environmental data (logs, assets, alerts) that represents the production environment.

Develop Grounded QA Dataset

Create question-answer pairs where the answer requires retrieving and synthesizing information from the data snapshot (e.g., "What user accounts logged into server 'X' during the time of alert 'Y'?").

Build Benchmark Harness Harness

Develop a testing framework that provides the LLM access to the relevant data snapshot when answering these questions.

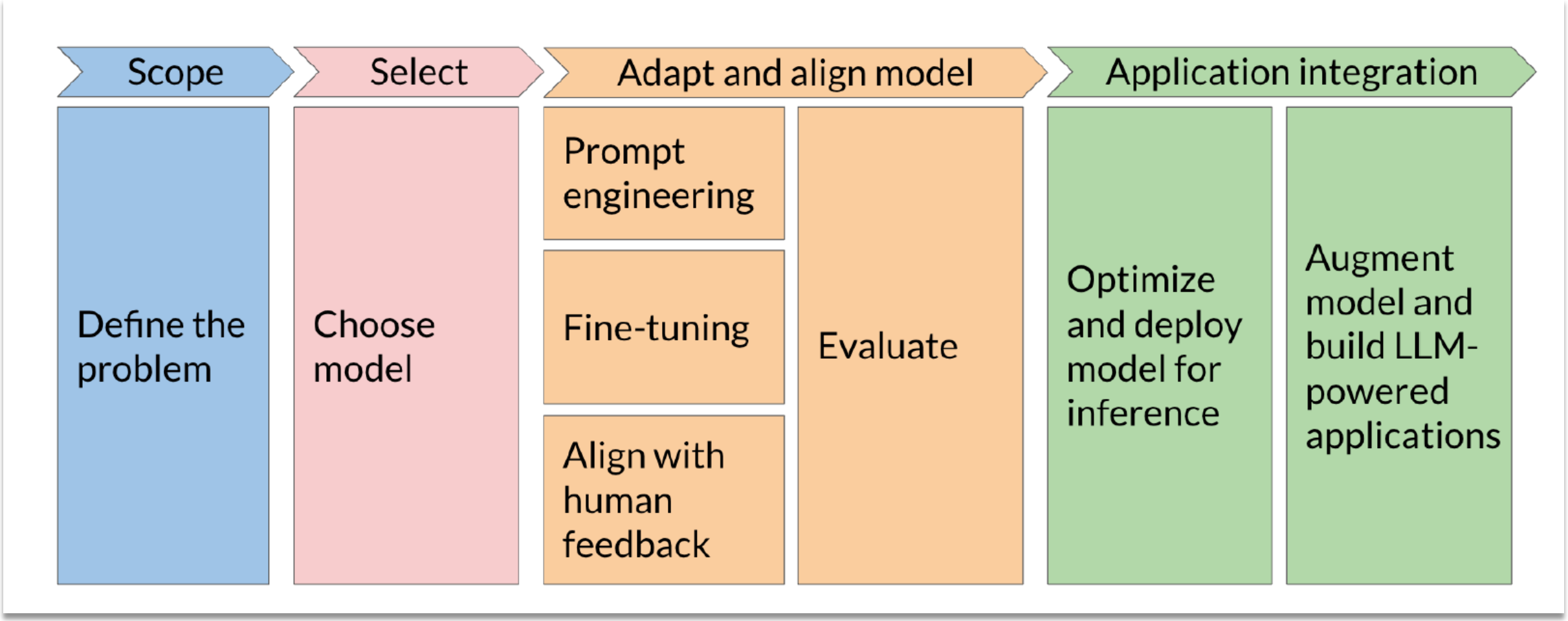
Evaluate Answer Quality

Assess answer correctness, completeness, and faithfulness to to the source data (checking for hallucination).

Standard benchmarks often test generic knowledge. The platform aims to answer questions grounded in the specific ingested environmental context (logs, environmental context (logs, assets, alerts).

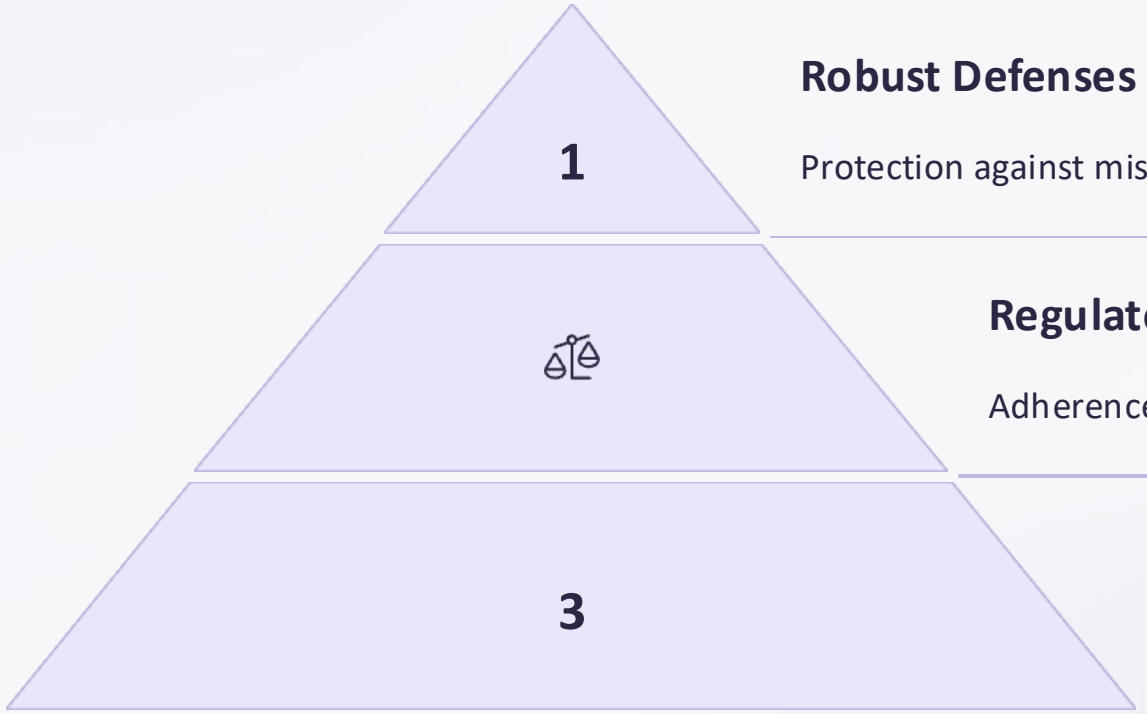
This requires significant effort but is crucial for evaluating the core RAG capability of the system – its ability to reason over specific environmental context.

Project Lifecycle



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Safety, Compliance, Red-Team



Robust Defenses

Protection against misuse

Regulatory Compliance

Adherence to legal requirements

Adversarial Testing

Proactive security validation

Ensuring the platform operates safely, securely, and in compliance with regulations is paramount, especially when dealing with sensitive data sensitive data and powerful LLMs.

Prompt Injection Defenses

Input Filtering/Sanitization

Attempt to detect and remove malicious instructions from user input (difficult to make foolproof).

Instruction Fine-tuning

Train the model on examples where it should should refuse harmful or out-of-scope requests. requests.

Output Filtering

Scan the LLM's generated response for harmful content, policy violations, or indicators of successful injection before showing it to the user.

Prompt Engineering

Structure prompts carefully to clearly delineate user input from system system instructions (e.g., using delimiters, XML tags).

Input Classification

Employ a smaller, fine-tuned model to classify user input intention before passing it to the main LLM.

LLMs are vulnerable to prompt injection, where malicious user input tricks the model into ignoring its instructions or perfor ming unintended actions.

Recommendation: Employ a multi-layered approach. Combine robust prompt engineering, instruction fine-tuning for refusal on common attacks, and output filtering as a final safety net. Relying on input filtering alone is generally insufficient.

Ongoing monitoring and red-teaming are needed to discover new injection vectors.

Audit Logging

Comprehensive Logging

Log every user prompt and the corresponding LLM response. response. Include metadata such as user ID, timestamp, timestamp, session ID, model version used, data sources sources consulted (retrieval context), and any safety flags flags triggered.

Secure Storage

Store audit logs securely in a dedicated, immutable logging system (e.g., write-once storage, dedicated security logging platform).

Retention Policy

Retain audit logs according to relevant compliance requirements and internal security policies (often longer than operational data retention).

Access Control

Implement strict access control on audit logs, limiting access to access to authorized security and audit personnel.

Comprehensive audit logging is essential for security monitoring, forensic review, and compliance.

Ensure the logging mechanism captures sufficient detail to reconstruct interactions and investigate potential misuse or security incidents.

Data Retention Policy

Hot Storage (30-90 days)

Active analytics database for immediate query access. Supports real-time analysis and recent investigations.

Warm Storage (Up to 1 year)

Cheaper disk or cloud storage (e.g., S3 Standard) for less frequent access. Supports extended investigations and compliance requirements.

Cold Storage (Remainder of retention retention period)

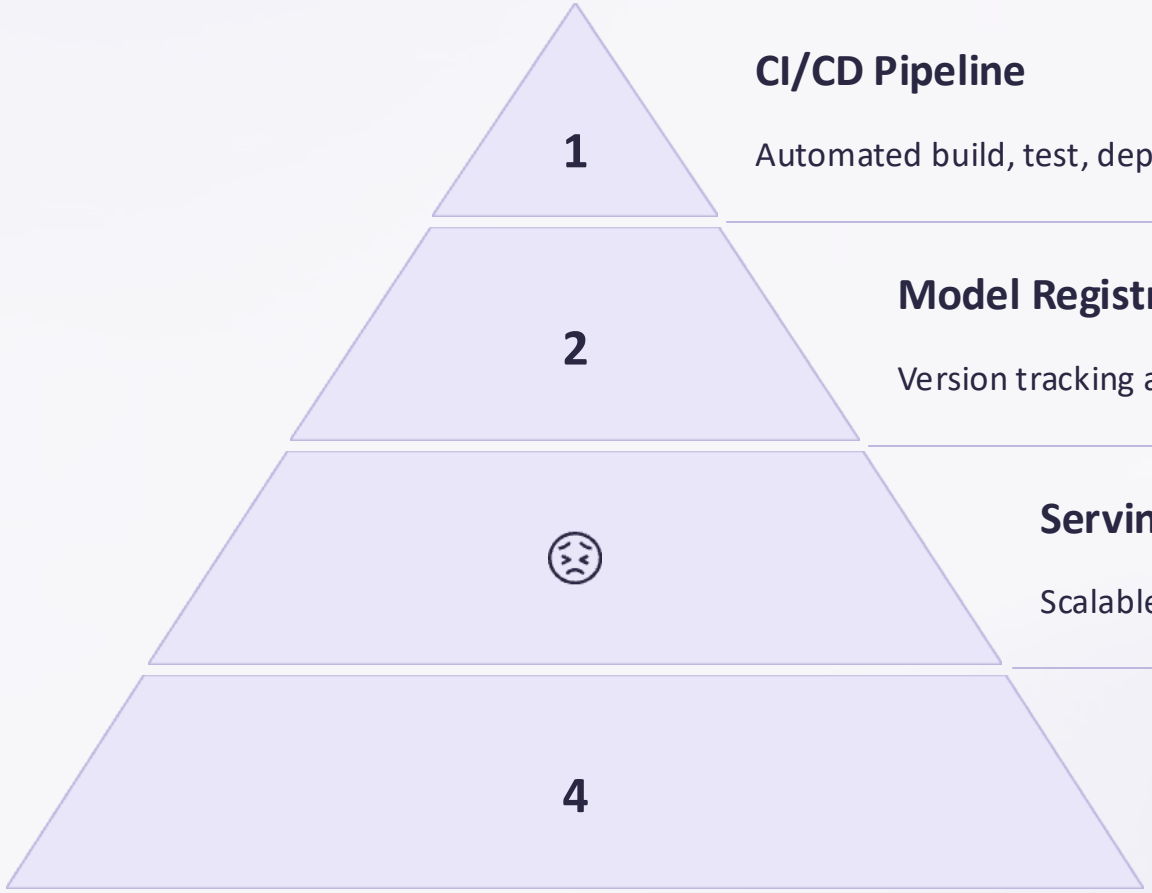
Archival storage (e.g., S3 Glacier Deep Archive, Azure Archive Storage) for long-term compliance. Minimal access expected.

Defining data retention periods is crucial for compliance, operational needs, and cost management. Policies must differentiate between raw and between raw and processed data.

Implementing tiered storage aligned with retention policies is essential for cost optimization. TTL (Time-To-Live) mechanisms within databases can within databases can automate the deletion of data from active tiers, while separate archival processes handle moving data to cold storage. storage.

The data retention policy requires formal sign-off from stakeholders, typically including the CISO, Legal/Compliance department, and potentially a and potentially a Data Governance Committee.

MLOps / Deployment



CI/CD Pipeline

Automated build, test, deploy

Model Registry

Version tracking and management

Serving Infrastructure

Scalable inference deployment

Monitoring & Feedback

Performance tracking and improvement

Efficiently developing, deploying, and managing the ML models and the overall platform requires robust MLOps practices and a well-defined deployment defined deployment strategy.

Serving Path Comparison

Triton Inference Server (NVIDIA)

High-performance inference server supporting multiple frameworks (TensorFlow, PyTorch, TensorRT) and backends. Offers features like dynamic batching, model ensembling, and GPU optimization. Can be complex to configure.

vLLM

An open-source library specifically specifically designed for fast LLM inference and serving. Uses techniques like PagedAttention to to optimize memory usage and throughput for LLMs. Gaining popularity for its performance.

Ollama

Focuses on ease of use for running LLMs locally or deploying them simply. May lack some advanced performance optimization or production features compared to Triton or vLLM.

Custom FastAPI/Flask + Transformers

Building a custom serving layer using using web frameworks and libraries libraries like Hugging Face Transformers. Offers maximum flexibility but requires implementing implementing features like batching, batching, queuing, and scaling manually.

Serving the fine-tuned LLM efficiently and reliably requires a dedicated inference serving platform.

Recommendation: For high-throughput, low-latency LLM serving, vLLM is a strong contender due to its specific optimizations for LLMs. Triton is a more general-purpose, mature option suitable if serving diverse model types is required.

Evaluate based on performance benchmarks with the chosen fine-tuned model and GPU hardware. Consider if GPU multi-tenant isolation is needed if multiple models or if multiple models or tenants share GPUs, which Triton might handle more robustly.

Rollout Policy and Rollback

Blue-Green Deployment

Maintain two identical production environments (Blue and Green). Deploy Deploy the new version to the inactive inactive environment (Green), test it, then then switch traffic.

Allows instant rollback by switching traffic back to Blue. Requires double the infrastructure resources.

Canary Rollout

Gradually route a small percentage of traffic to the new version. Monitor performance and error rates. Increase traffic progressively if stable.

Allows detecting issues with minimal user impact. More complex traffic routing logic required.

Rollback Trigger

Define automated triggers for rollback rollback based on key metrics: increase in increase in error rate, latency spike, drop drop in benchmark performance (from (from real-time evaluation or shadow deployment), critical safety/alignment safety/alignment failures.

Deploying new model versions into production requires a safe rollout strategy.

Recommendation: Canary rollouts are generally preferred for ML models, as they allow gradual exposure and monitoring of subtle performance changes or unexpected behaviors with real traffic.

Combine with automated monitoring and clear rollback triggers based on critical KPIs (latency, error rate, core task accuracy).

Summary Value Proposition

- Industry Fragmentation yields opportunity with Integrated Approach
- Consolidated and normalized threat identification across the entire network infrastructure (Firewall, WAF, SIEM, NIDS, EDR)
- Build and train CTI-LLM using state of the art threat intelligence (Flashpoint's Ignite)
- RAG based integration into CTI-LLM for automated contextualized responses
- Yield better outcomes both in terms of threat detection and response, a necessary condition for Cyber Defense in the age of AI

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